Never-Realized Capital Gains

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First version: November 2024
This version: August 2026
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In realization-based tax systems, capital gains are not taxed until the asset is sold. Policies of "stepped-up basis" make capital gains deferral permanent when appreciated assets are inherited. We construct novel measures of unrealized and inherited capital gains in Norway. Unrealized gains are large, top-heavy, and poorly proxied by realized gains. Gains passing through intergenerational transfers equal 57% of the realized gains tax base. Two difference-in-differences designs show Norway's 2006 removal of stepped-up basis raised capital gains tax revenue by 26%. Yet realizations are small relative to accrual, so the stock of unrealized gains continues to grow after the reform.

JEL: H20, H31, D31, G51

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Msall wishes to thank her advisors Marianne Bertrand, Mikhail Golosov, Magne Mogstad, and Eric Zwick for all their support. We also thank Kristoffer Berg, Eric Budish, Nathan Hendren, Valerie Michelman, Matthew Notowidigdo, Ben Sprung-Keyser, Kjetil Storesletten, and many seminar participants for helpful discussions and feedback. We thank Peter Birch Sørensen and Frederik Zimmer for valuable contemporary insight into the policy changes studied in this paper. We thank Lasse Eika, Ola Vestad and Arnstein Vestre for help accessing and understanding the Norwegian administrative data. Msall gratefully acknowledges support from the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE-1746045. Any opinions, findings, and conclusions, or recommendations expressed in this material are those of the authors and do not represent the views of the National Science Foundation. All errors are our own. Msall: lucymsall@uchicago.edu. Næss: oleandreas.naess@gmail.com.

Over the past decade, numerous proposals have sought to expand or reform the taxation of wealth and capital. These proposals are often motivated by the concern that income tax systems leave many capital gains permanently untaxed. In the US, UK, and ten other OECD countries, this concern is exemplified by "stepped-up basis" policies, in which capital gains left in intergenerational transfers (rather than realized in a sale by their original owner) are exempted from the income tax base.[1] Debate on these issues is divided over whether unrealized capital gains constitute a meaningful gap in the tax base, and, if so, whether the problem is stepped-up basis specifically or features of realization-based income taxation more generally. Several primitives are central to this debate and largely previously unmeasured: the magnitude of unrealized and never-realized capital gains, the distribution of such gains across assets and owners, and the extent to which stepped-up basis causally inhibits realization.

This paper uses high-quality administrative data from Norway to develop a novel approach to measuring unrealized gains and examine a rare policy experiment that removed stepped-up basis. We address two sets of questions. First, descriptively: how large is the stock of unrealized capital gains and how much passes across generations without realization by the original owner? Second, causally: how does removing stepped-up basis affect realizations and intergenerational transfers, and how large are these responses relative to the accumulated stock of unrealized capital gains?

Two key challenges have inhibited prior empirical study of these questions. The first challenge is that, globally, tax records often include information on realized capital gains, but generally do not measure unrealized gains or the gains embedded in gifts and bequests. As a result, little is known about which gains remain outside the realized tax base, or who benefits from stepped-up basis. We overcome this challenge by reconstructing detailed capital gains records for assets held by Norwegian households. We link Norwegian administrative data on household wealth and income, asset purchases and sales, and current asset values at the individual-asset-year level. Although Norwegian tax records only measure realized capital gains (RCG), our linked data allow us to construct measures of unrealized capital gains (UCG) and inherited capital gains (ICG) in listed stock, private businesses, and real estate holdings.

The second challenge is policy variation. Despite many proposals to modify the tax treatment of inherited capital gains, few policy changes have actually been enacted in a way that allows for

[1] Chile, France, Korea, Lithuania, Portugal, Slovenia, Spain, the United Kingdom, and the United States all apply step-up to most deathtime inheritance transfers; while Denmark, Finland, and Hungary have more complicated regimes in which step-up applies to only some types of inherited assets (OECD, 2021). Most countries do not apply step-up to capital gains passed as *inter-vivos* gifts.

empirical evaluation of the resulting behavioral effects.[2] In this paper, we overcome this challenge by studying Norway, the only country we are aware of that has permanently and substantially modified its treatment of inherited capital gains in the last four decades by removing stepped-up basis.

In the first part of the paper, we establish four facts about unrealized and inherited capital gains in Norway prior to the removal of stepped-up basis. First, unrealized capital gains are large and important for understanding top wealth: 18–20% of total household wealth is composed of unrealized capital gains in assets subject to the capital gains tax, and that fraction rises above 50% for the wealthiest households. Second, realized capital gains are not representative of the broader stock of gains. Relative to realized gains, unrealized gains are more concentrated at the top of the wealth distribution and more heavily embedded in private businesses. For example, unlisted stock accounts for 82% of the observed stock of UCG in our taxable asset categories, but only 42% of observed RCG in the same categories. Standard tax data on realized gains therefore make capital gains look more liquid, less private-business-intensive, and less top-heavy than the underlying stock of unrealized gains. Third, step-up exempts a large flow of gains embedded in highly appreciated inherited assets. In 2004, the last full year before step-up's removal was announced, inherited capital gains (exempted from tax under stepped-up basis) equaled 57% of the annual realized capital gains tax base. This large exemption arises because assets transferred to heirs contain far more embedded appreciation on average than assets their owners choose to sell: the ratio of capital gains to value is 56% for inherited taxable assets, compared to 8% for realized taxable assets. Fourth, the gains exempted by step-up accrue overwhelmingly to the top of the wealth distribution. The top 1% of households are the source of more than half of inherited capital gains in taxable assets, and the top 0.1% alone is the source of roughly thirty percent.

In the second part of the paper, we examine how the removal of step-up in Norway affected realization behavior. Prior to 2006, Norway allowed accrued capital gains to disappear from the tax base when appreciated assets were gifted or bequeathed (a policy of "stepped-up basis" for both *inter-vivos* gifts and testamentary bequests).[3] In 2006, Norway began "carrying over" the

[2] The United States briefly adopted carryover basis in 2010, but estates of Americans who died that year were ultimately allowed to choose between paying the estate tax and receiving stepped-up basis, or receiving carryover basis treatment. Gordon et al. (2016a,b) evaluate what may be learned from this brief, "voluntary" tax regime. The US also passed but never implemented a repeal of step-up in favor of carryover basis in 1976.

[3] Throughout this paper, we use the terms "inherited assets" and "inherited capital gains" to refer to assets or capital gains received in intergenerational transfers regardless of whether the transfer occurred during the older generation's lifetime or at their death. We occasionally distinguish the former as "*inter-vivos* transfers," and call the latter "deathtime" transfers.

original owner's unrealized capital gains to recipients of stock (including private business interests and mutual fund shares) transferred as gifts or bequests. The reform did not change the capital gains tax rate faced by the original owner, but instead increased the future tax cost of leaving appreciated stock unrealized for intergenerational transfer. We present a conceptual framework showing that exposure to this cost increase is greater for older investors and for investors with a larger ratio of unrealized gains to wealth.

Our empirical strategies exploit these dimensions of differential exposure to the reform. We first estimate difference-in-difference regression models that compare the effect of the reform on the realizations of individuals aged 30–49 versus those of older individuals. The identifying assumption for this strategy is that, absent the reform, different age groups' realized capital gains would have the same percent change reactions to yearly shocks (i.e., that there would counterfactually be a stable relative relationship between age and log average RCG). We find that investors in older age groups responded to the removal of step-up by increasing their taxable realizations relative to this counterfactual. Investors aged 50 and above increased yearly realized capital gains by \$314 (32%) on average after the reform announcement, with response magnitudes that increase by age. As a second, alternative, empirical strategy, we compare realization responses by portfolio composition for investors of similar age and wealth. We estimate difference-in-difference regression models that compare the effect of the reform for investors with different levels of unrealized capital gains in stock prior to the reform. The identifying assumption is that, absent the reform, investors with different unrealized capital gains in stock but the same age and wealth level would have capital gains realizations (of any asset type) that trend with the same percent changes over time. Results from this design imply a similar aggregate increase in realized capital gains among older investors as in the first design. Increases in taxable realizations were largest among the investors with the most highly appreciated stock portfolios. The announcement of step-up's impending removal also triggered a large anticipatory wave of *inter-vivos* transfers, but the simultaneous increase in taxable realizations shows that early transfers were an imperfect substitute for retaining appreciated assets.

Combining the two parts of the paper produces findings that cut in two directions. The removal of step-up has a large effect on the Norwegian capital gains tax base, raising annual capital gains tax revenue by 26% relative to counterfactual. However, the additional realizations only amount to 0.5% of the 2004 stock of unrealized capital gains per year, while new gains accrue at a much faster rate. Even after step-up is removed, the stock of unrealized capital gains continues to grow throughout our post-period. Removing step-up therefore captures revenue from gains that would otherwise have been inherited tax-free, but the stock of unrealized appreciation is so large that the reform does not materially reduce it.

Step-up policy has been subject to debate in both the US and UK. Our results clarify that step-up benefits very wealthy households and that, relative to other policies that would benefit a similar group, step-up creates a negative fiscal externality. The reason is that step-up increases the return to deferral and thereby amplifies the lock-in distortion created by realization-based capital gains taxation. A budget-neutral reform that repealed step-up and returned the revenue as a lower capital gains tax rate would thus benefit a similar, if somewhat broader and less wealthy, group of capital gains owners while reducing rather than amplifying distortions.

Our paper contributes to two literatures. First, our data work complements a recent literature which considers whether to incorporate yearly flows of accruing capital gains, asset returns or retained business income into a broader concept of personal income (Thoresen et al., 2012; Wolfson et al., 2016; Fairfield and Jorratt De Luis, 2016; Eika et al., 2020; Larrimore et al., 2021; Yagan, 2023; Aaberge et al., 2024; Fagereng et al., 2025; Alstadsæter et al., 2026; Ring et al., 2026). Because prior papers do not link owners' original purchase prices to their contemporary holdings, they cannot measure the accumulated stock of unrealized gains or the gains embedded in intergenerational transfers. By reconstructing basis at the individual-asset level, we provide, to our knowledge, the first asset-level measures of unrealized and inherited capital gains in non-survey data.[4] Second, we add to a long literature on behavioral responses to capital gains taxation (e.g., Feldstein and Yitzhaki (1978)). Much of this literature estimates how realized capital gains respond to changes in the capital gains tax rate.[5] We study realization responses to a different policy lever: changes to the tax treatment of inherited capital gains, holding the statutory rate fixed.[6] We find this policy lever to be comparatively powerful: to produce a similar realization response through a rate cut, we estimate (in a back-of-the-envelope calculation using elasticities from the recent literature) that Norway would have had to cut its capital gains tax rate by 7-19 percentage points. On the other hand, we contrast this "large" realizations response (the typical focus of the literature) with the small effect on the overall stock of unrealized capital gains.

[4] The US Survey of Consumer Finances (SCF) measures unrealized capital gains but, by design, excludes the very wealthiest US households, preventing a complete characterization of the distribution of unrealized capital gains. The SCF's cross-sectional structure also makes it difficult to use to analyze behavioral changes over time.

[5] Recent estimates of the response of realized capital gains to the capital gains tax rate include Agersnap and Zidar (2021), Dowd and McClelland (2024) and Lavecchia and Tazhitdinova (2024); see Gravelle (2026) for a detailed review of this literature.

[6] Although we provide the first quasi-experimental evidence examining a permanent change to the tax treatment of inherited capital gains, we are certainly not the first to theorize that step-up likely depresses capital gains realizations (Holt and Shelton, 1962; Auerbach, 1989; Kiefer, 1990).

1 Data and Measurement

Our paper uses data and policy variation from Norway, primarily because Norway is the only country that has enacted a permanent change to its stepped-up basis policy in recent decades. Norway is also a uniquely appropriate setting for this project because the Norwegian Tax Authority collects rich data on taxable income, wealth, and asset transactions.[7] The core data contribution of this paper is an asset-level reconstruction of the capital gains universe. Throughout the paper, we refer to the stock of unrealized capital gains as UCG, and the flows of realized and inherited capital gains as RCG and ICG, respectively. Norwegian tax returns directly measure RCG, but do not measure the stock of UCG or the ICG embedded in assets transferred to heirs. We construct these measures by linking transaction records, purchase prices and current asset values at the individual-asset-year level. Appendix Table A1 summarizes the role each data source plays in our measurement exercise. We convert all currency amounts to 2018 USD using yearly average consumer price indices from Statistics Norway and the NOK-USD exchange rate on December 31, 2018.

1.1 Administrative Data Sources

Our measurement exercise combines two broad types of Norwegian administrative data: annual individual tax records, and asset-level transaction records. As in many countries, Norwegian taxpayers report taxable RCG as a component of income on their yearly tax returns. We use this RCG variable as our primary outcome variable in our regressions in Section 5. For a subset of years, we can decompose total taxable RCG into taxable RCG from real estate, from unlisted stock and from other asset types (primarily listed stock and mutual funds).[8] The annual tax records also include detailed information about each Norwegian household's net wealth. We are confident that the wealth data capture market values well for most components of financial wealth, such as bank deposits, liabilities, and listed securities.[9] For unlisted companies we replace the tax-assessed values with our estimates of market value described in Section 1.2.2. We also replace tax-assessed

[7]As background, Norway is a small, open economy with a well-educated population, a sizable middle class, and a significant public sector. The Norwegian population is 5.4 million people and consistently ranks among the wealthiest global populations. Levels of wealth inequality are relatively high, and top wealth is concentrated in real estate and private business interests (see Appendix A).

[8]The tax returns allow us to decompose aggregate RCG into real estate and non-real estate RCG for all years 1998–2018 except for 2005 (due to a data corruption). Once the stock transactions data starts in 2003, we can further decompose non-real estate RCG into RCG from and not from particular kinds of stock.

[9]Norwegian administrative data is considered very high quality and nearly all data on income and financial wealth is third-party reported from employers, banks, and financial intermediaries.

real estate values with property-level values computed by Eika et al. (2020).[10]

Since the Norwegian Tax Authority does not collect data on UCG or ICG, we use asset-level transaction data to build these measures ourselves for the asset categories of Norwegian stock (including private businesses) and real estate. We observe individual transactions, including sales, inheritance transfers and other transaction types, for listed and unlisted stocks[11] from 2003 to 2018. These data contain detailed information on the type of transfer, the identity of the buyer or seller, and the transaction price. The data contain information on the last transaction prior to January 1, 2003, and all subsequent transactions from 2003 to 2018. We have similar real estate transaction data covering 1993 to 2018 (including the last transaction prior to 1993) from the Norwegian Land Register. The pre-period last transaction information allows us to initialize holdings and basis for assets already held when the panel begins. For example, if an individual bought shares in 1985 and continued to hold them in 2003, we observe both the quantity held and the original purchase cost.

1.2 Constructing Asset-Level UCG and ICG

At the individual-asset level, capital gains are measured as the current value of the asset minus the owner's "basis" in the asset (typically the owner's original purchase cost). That is, Capital Gains = Current Value - Basis. For RCG, current value in this expression is observed directly as the transaction price when the asset is sold or otherwise realized. The measurement challenge in this paper is that an analogous current value is not observed when an asset remains unsold at a particular point in time, or when it is transferred to heirs without an arm's-length sale price. We therefore need to reconstruct these values for our UCG and ICG estimates.

1.2.1 Tracking Basis and Current Holdings

To construct our capital gains measures, we first convert the transactions data into a ledger at the individual-asset level. When an individual buys a new asset, we open a ledger for that individual-asset pair. We record the number of shares initially owned in the asset (or fractional ownership share for real estate) and then update the number of shares held (or fractional ownership held) as the individual makes subsequent transactions in the asset. We account for stock splits, consolidations

[10] Although the Norwegian tax authority assesses the value of all domestic real property each year, the quality of the assessment methodology prior to 2010 is thought to be poor, and the 2010 revision did not improve the quality of the assessments for non-residential property (Ring, 2025). The Eika et al. (2020) values we use for real estate valuation are built from the same real estate transaction data we use to track ownership by interpolating and extrapolating from sale values with house price indices in years when the specific property does not sell.

[11] Our category of unlisted stock includes all private business interests other than sole proprietorships.

and mergers when applicable.

We measure an individual's initial basis in an asset as the asset's original purchase cost, or carryover basis if applicable. To properly account for carryover basis, we must trace assets moving across ledgers in non-taxable transactions such as mergers, changes to share classes, or intergenerational transfers after the removal of step-up (The Norwegian Tax Administration, n.d.). As the individual buys and sells, we increase and decrease his basis under a first-in-first-out assumption, in keeping with the accounting methods required by Norwegian law.[12]

1.2.2 Measuring Current Value for Unrealized and Inherited Capital Gains

To construct UCG, we require a yearly measure of each asset's market value, even when that asset is not sold. For listed stock, we obtain the stock's end-of-year closing price from Thomson-Reuters Datastream. For unlisted stock, we begin with the book value measure used by the Norwegian government to value each unlisted company for wealth and inheritance tax purposes.[13] Valuing private businesses is inherently difficult, and tax-assessed book values are likely to understate market values for many unlisted firms. We therefore create our own measure of unlisted company value by scaling up the company's sales, EBITDA, and non-financial assets with multipliers computed for a sample of listed firms in the same industry, and then applying a 10% liquidity discount to those values. This procedure follows Smith et al. (2023) and is described in greater detail in Appendix D.2. When a firm's estimated value is negative, we replace that value with zero, in keeping with the limited liability of the firms considered. When unlisted firm value is directly attributable to firm ownership in another firm or listed equity, we value that portion of the firm as the pro-rata share of the child firm's value or using the value of the listed equity. As our yearly measure of real estate value, we again use the property-level values computed by Eika et al. (2020).

We observe gift and bequest transfers of stock and real estate. To construct our measure of ICG, we assign the relevant portion of the predecessor's basis to the transferred units and calculate ICG as the value of the transferred asset minus the basis associated with the asset. For listed stock, we value transferred shares using observed market prices at the time of transfer. For unlisted stock and real estate, we use the asset's estimated value at the most recent year end.

[12] Appendix Section D.3 shows that our measure of UCG is qualitatively unaffected if we instead assume capital gains are realized according to an average basis assumption.

[13] When stepped-up basis was in place, this valuation also served as the heir's new basis for unlisted stock received in inheritance.

1.2.3 Validation

Our capital gains measures rely on reconstructing holdings and basis from transactions. Appendix Section D validates our ledger-implied holdings against end-of-year shareholder reports in the Norwegian Shareholder Registry, reported by limited liability companies since 2004, and finds that our ledger agrees very closely with the Registry: an exact match for 97% of current shareholders. We also validate our measure of basis using realized transactions. For capital gains that are eventually realized, we can calculate realized capital gains as the recorded value of the sale minus basis and compare that calculation to the realized capital gains reported in each individual's tax return. This match is also good — 97% of individual yearly tax returns agree, within rounding error, with our calculation for the amount of realized capital gains in stock (see Appendix D).

1.3 Coverage and Limitations

We focus on capital gains in what we call *taxable asset categories* — that is, asset categories that are usually subject to the capital gains tax. Taxable asset categories in our data include listed and unlisted stock, mutual funds, and taxable real estate. Primary residences and certain other forms of real estate are exempt from capital gains tax if the owner meets certain holding period requirements.[14] These categories of *non-taxable real estate* do not typically benefit from stepped-up basis; we consider their capital gains separately in the analysis that follows.

We construct UCG and ICG for Norwegian stock (listed and unlisted) and real estate. A limitation of the transaction data is that we do not have access to transactions in mutual funds, which we know from aggregate statistics to be responsible for nearly all RCG that are not from directly owned stock or real estate (Statistics Norway, 2007). Since we observe wealth in mutual funds but cannot calculate their UCG, we bound our estimates of total UCG by making the alternative assumptions that (1) Norwegian households do not have any UCG in mutual funds and (2) Norwegian households' mutual fund wealth is 100% UCG, i.e., all owners of mutual funds have zero basis in their mutual fund holding. In practice, the treatment of mutual funds does not qualitatively affect any of our results. While mutual funds are an important component of many Norwegians' portfolios, they are much less important for the richest Norwegians (see Appendix Figures A6 and A7). We do not directly measure UCG in other asset classes, such as directly held international stock, directly held business assets, and artwork. However, the aggregate amount of

[14] Appendix Section F details the holding period requirements for real estate exemptions from capital gains taxes.

wealth in these categories, and thus the upper bound for the possible amount of UCG, is small.[15] For our regression analyses, our primary outcome variable of RCG is taken directly from individual tax returns and so includes all taxable asset categories.

A few categories of wealth are missing. We do not observe wealth in defined contribution pension plans. Defined contribution plans were not permissible in Norway until 2001 and only begin to have market share during our period of study. Although Norwegians are legally required to report wealth both inside and outside Norway, we do not observe wealth that Norwegians fail to report to the tax authority. Alstadsæter et al. (2019) establish that offshoring forms of tax evasion are substantial among the wealthy citizens of Norway. Our inability to observe this offshore wealth may mean our estimates understate the amount of UCG in portfolios at the top of the wealth distribution or incorrectly rank evading households in the wealth distribution. On the other hand, since these assets are, by definition, hidden from the Norwegian tax system, we do not expect realizations or inheritance of these assets to affect our analysis of the step-up reform.

2 Step-Up in Context: The Magnitude and Distribution of Capital Gains

In this section, we use our new data to document the magnitude, composition and distribution of unrealized and inherited capital gains under step-up (i.e., prior to the Norwegian policy reforms). This section focuses on capital gains in Norway as of end-of-year 2004, immediately before the removal of step-up was announced.

Fact 1. Unrealized Capital Gains Are a Large Fraction of Household Wealth

UCG in taxable asset categories are large: eighteen to twenty percent of total household wealth. At the end of 2004, \$91.92–\$102.52 billion of UCG were outstanding in asset categories subject to the capital gains tax.[16] See Table 1 Panel (a). This amount equals 18–20% of aggregate net wealth held by Norwegian households in 2004 (\$500 billion).

Of the \$146.25 billion of wealth in the taxable asset categories of listed stock, unlisted stock & taxable real estate, 63% was UCG. Unlisted stock was particularly appreciated: 74% of wealth in unlisted stock was UCG, compared to 23% in listed stock and 41% in taxable real estate (Table

[15] Most Norwegians participate in international equity markets via mutual funds, and this wealth would therefore be included in our mutual fund numbers.

[16] We present our measures of total unrealized capital gains in taxable assets as a range to acknowledge our uncertainty regarding the amount of unrealized capital gains in mutual funds (see Section 1.3).

1 Panel (a)). These magnitudes establish the scale of the latent capital gains tax base: before considering which gains are eventually sold or transferred, accumulated untaxed appreciation is a quantitatively important component of household wealth.

Table 1: Unrealized, Realized, and Inherited Capital Gains in 2004 (billions of 2018 USD)

(a) Asset Value & Unrealized Capital Gains

Asset Type	Total Value	UCG	UCG/Value
Taxable Assets including Mutual Funds	156.85	102.52†	0.65†
Mutual Funds	10.60	-	-
Taxable Assets excluding Mutual Funds	146.25	91.92	0.63
Listed Shares	9.15	2.12	0.23
Unlisted Shares	101.00	75.10	0.74
Taxable Real Estate	36.10	14.70	0.41
Nontaxable Real Estate	348.00	141.00	0.41

(b) Realizations & Realized Capital Gains

Asset Type	Amount Realized	RCG	RCG/Amount
Taxable Assets excluding Mutual Funds	14.84	1.14	0.08
Listed Shares	12.30	0.31	0.03
Unlisted Shares	1.59	0.48	0.31
Taxable Real Estate	0.95	0.34	0.36
Nontaxable Real Estate	5.49	1.79	0.33

(c) Inheritances & Inherited Capital Gains

Asset Type	Amount Inherited	ICG	ICG/Amount
Taxable Assets excluding Mutual Funds	1.16	0.65	0.56
Listed Shares	0.05	0.03	0.66
Unlisted Shares	0.54	0.39	0.72
Taxable Real Estate	0.57	0.22	0.40
Nontaxable Real Estate	3.40	1.52	0.45

Note: This table presents aggregate numbers on the value of wealth and capital gains in certain asset categories in Norway at the end of 2004. Panel (a) presents the total amount of wealth ("Total Value") in each category, along with the amount of Unrealized Capital Gains ("UCG"). The amount of UCG when mutual funds are included assumes that mutual funds are 100% appreciated (estimates affected by this assumption are marked with a dagger †). Panel (b) displays the value of assets in these categories that are realized in 2004 ("Amount Realized"), along with the Realized Capital Gains ("RCG"). Panel (c) displays the value of assets in these categories that are inherited in 2004 ("Amount Inherited"), along with the Inherited Capital Gains ("ICG").

Fact 2. Realized Gains Are Unrepresentative of Unrealized Gains

Relative to the more commonly observable RCG, the UCG stock is more heavily tilted toward private businesses, and its distribution across households is more concentrated at the top of the wealth distribution. Listed shares account for 83% of realized asset value and RCG is attributable to a mix of asset classes.[17] On the other hand, unlisted stock is responsible for the majority of wealth in capital gains-producing assets, UCG and ICG. Table 1 shows that unlisted stock accounts for 69% of value in our taxable asset categories ($101.00 billion of the $146.25 billion), 82% of the UCG stock, 47% of inherited assets and 60% of inherited capital gains. By comparison, unlisted stock accounts for 42% of observed RCG and 11% of realized asset value.

Figure 1 Panel (a) compares the concentration of RCG and UCG across households ranked by total net worth. The UCG curve lies below the RCG curve through the upper part of the distribution, indicating that UCG are more concentrated at the top than RCG. Put differently, tax data on RCG make capital gains look less top-heavy than when unrealized capital gains are included. In 2004, the bottom 90% of households owned less than 14% of the stock of taxable UCG in Norway. The next 9% of households owned 23-24%, and the top 1% of households owned 63-68% of the stock of UCG.[18]

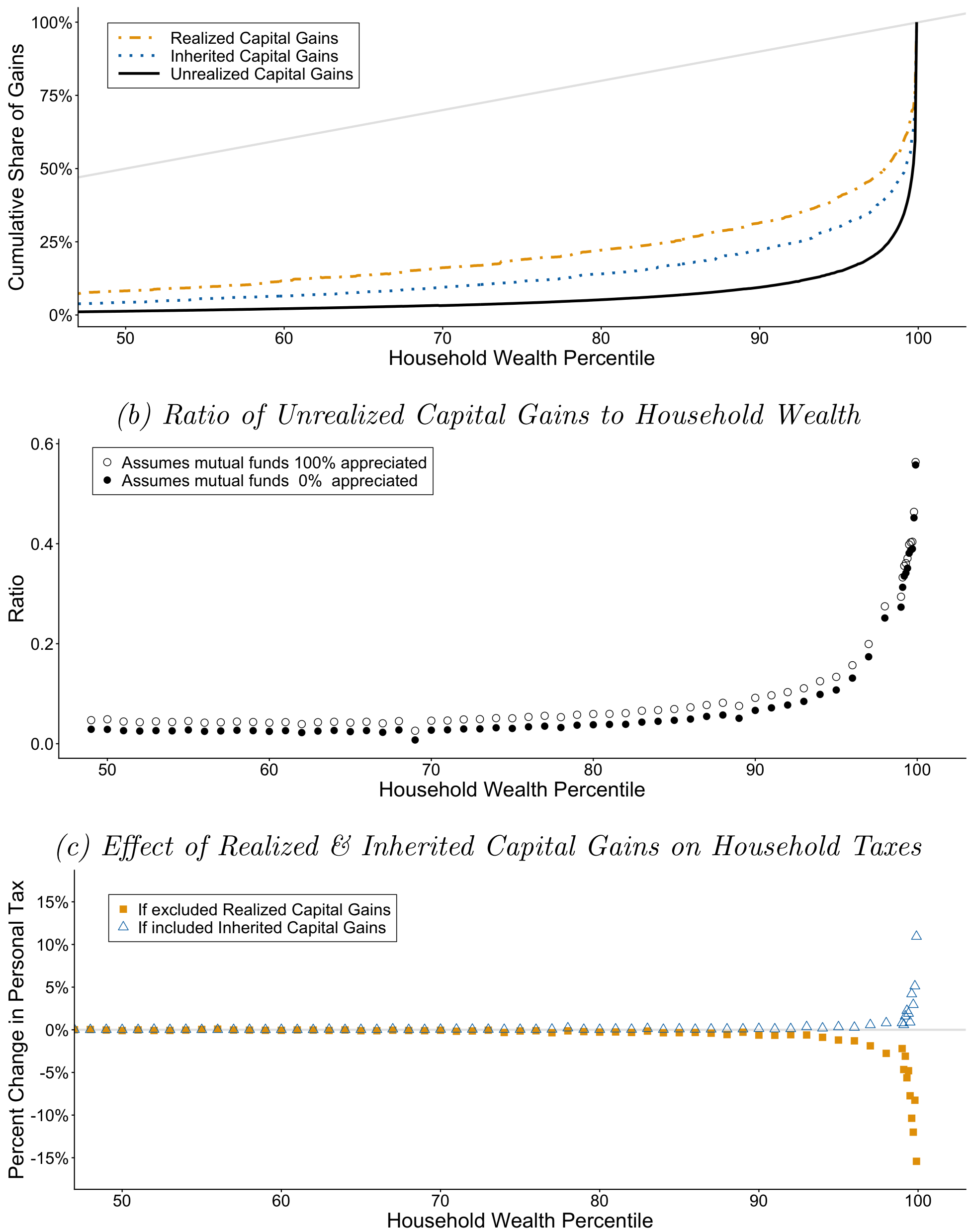
The importance of UCG within household portfolios also varies across the wealth distribution. Figure 1 Panel (b) shows that taxable UCG are less than 10% of wealth for the bottom 90% of households in 2004. The ratio of taxable UCG to total household wealth increases sharply at the top of the wealth distribution, reaching a value above 50% for the wealthiest households.

The skewed distribution of taxable UCG is mostly driven by differences in household portfolio composition across the wealth distribution. Wealthy households hold large fractions of their wealth in assets subject to the capital gains tax — holdings of unlisted stock are particularly important for the wealthiest households. For example, households in the top 1% of net wealth in 2004 own 62% of the total wealth in stock and taxable real estate (see Appendix Figure A7). These top households claim a similar share of unrealized capital gains in these asset categories. In contrast, less wealthy households own more of their wealth in cash and primary residences, which are not subject to the capital gains tax.

[17] Although listed shares make up 83% of realized asset value, realizations of listed stock frequently contain losses or very little capital gain. Although unlisted stock is realized less frequently, realizations of unlisted stock tend to be much more appreciated (a 31% RCG/Amount ratio, versus 3% for listed stock realizations) and therefore make up a larger share of RCG than of realized asset value.

[18] As before, we present ranges to incorporate alternative assumptions regarding the amount of unrealized capital gains within mutual funds.

Figure 1: Capital Gains Across the Household Wealth Distribution, 2004



(a) Distribution of Capital Gains, by Type

(b) Ratio of Unrealized Capital Gains to Household Wealth

(c) Effect of Realized & Inherited Capital Gains on Household Taxes

Note: This figure depicts how capital gains vary across households in 2004. Households are grouped on the x-axis by their percentile of total net wealth in 2004. Panel (a) compares the concentration of realized, inherited, and unrealized capital gains by plotting the proportion of gains held by the bottom x% of households, where households are ranked by their total net worth (a la Lorenz (1905)). The light gray line depicts perfect equality ($y = x$). Unrealized and inherited capital gains are significantly more concentrated at the top of the wealth distribution than the distribution of realized capital gains. Panel (b) depicts average ratios of unrealized capital gains in assets subject to the capital gains tax (mutual funds, listed and unlisted stock and taxable real estate) to total household net worth. Panel (c) quantifies the mechanical average effect that either excluding realized capital gains from the tax base or including inherited capital gains in the giver's tax base (at the regular capital gains rate) would have on the personal tax burdens (i.e., income + wealth taxes) of households ranked by total net worth. Note that Panels (a) and (c) use the lower-bound (0%) assumption for mutual fund appreciation and that inherited capital gains are attributed to the gifting household.

Fact 3. Step-Up Exemptions are Large Relative to the Tax Base but Small Relative to the UCG Stock

In 2004, \$650 million of capital gains in stock and taxable real estate were passed through inheritance and thus exempted from tax under the stepped-up basis policy. In the same year, \$1.14 billion of capital gains were realized and taxed in stock and taxable real estate (Table 1 column 2). Therefore, Norway's stepped-up basis policy exempted an amount equal to 57% of the annual capital gains tax base.

Step-up exempts a large fraction of the capital gains tax base despite applying to a small volume of transfers: inheritance transactions contained only one-thirteenth of the asset value of realization transactions (Table 1, Panels (b) and (c), column 1). But there are large differences in appreciation between inherited and realized assets. The ratio of taxable capital gains to value in inherited assets subject to the capital gains tax is 56%. Realized assets are much less appreciated: the ratio of capital gains to value in realized assets subject to the capital gains tax is 8% (Table 1, Panels (b) and (c), column 3).

Relative to the capital gains tax base (RCG), step-up is large, so removing step-up could be expected to have a substantial mechanical effect on capital gains tax collections. But relative to the stock of UCG, both RCG and ICG are small yearly outflows: in 2004, ICG in stock and taxable real estate equaled 0.7% of the UCG stock in those categories (\$92 billion), while RCG equaled 1.2% of the same stock. These benchmarks therefore reveal a nuanced picture of step-up's significance.

Fact 4. The Gains Exempted by Step-Up Come Overwhelmingly from the Top

The ICG exempted by step-up are highly concentrated at the top of the wealth distribution. Figure 1 Panel (a) shows that the bottom 90% of households are the source of 22% of ICG in taxable assets in 2004. The next 9% of households are responsible for 25% of ICG, while the top 1% is the source of the remaining 53%. The top 0.1% alone is responsible for 30% of ICG in taxable assets.

Figure 1 Panel (c) translates these distributional facts into their implications for personal tax burdens. We calculate the mechanical change in average personal taxes across the wealth distribution under two counterfactual scenarios: one in which RCG are exempted from tax, and one in which ICG are also taxed at the 28% capital gains tax rate. Both capital gains categories have a near-zero effect on personal taxes for most of the wealth distribution. But in the upper percentiles, removing step-up would, in the absence of behavioral responses, increase personal taxes by approximately 11% for the very wealthiest households. This is similar in size to the approximately 15% decrease in personal taxes at the top that would result from exempting all RCG. The comparison also illustrates the relative concentration of ICG versus RCG: the taxation of RCG affects a somewhat broader swath of the top 10%, while ICG only matter above the 99th percentile.

3 Norway's Removal of Step-Up for Stock (2005-2006)

The remainder of this paper studies the removal of stepped-up basis for stock in Norway (including listed and unlisted stock, and equity-based mutual fund shares). This change was announced in 2005 and implemented in 2006. During most of our period of study, capital gains realized by their original owner were taxed at a 28% rate.[19] Heirs were subject to an inheritance tax on the total value of received gifts and bequests (of any type), but, until 2006, were not subject to capital gains tax on gains accrued during their predecessor's holding period. The top inheritance tax rate for transfers between close relatives was 20% from 1999 to 2008.[20] We end our short-term difference-in-difference results in 2008 to isolate the removal of step-up for stock from subsequent changes to inheritance tax rates (beginning in 2009) and capital gains tax rates (beginning in 2014).[21]

The repeal of step-up was enacted as a part of the broader 2006 tax reform mostly focused on dividend taxation. Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until December 2004 (Norwegian Ministry of Finance, 2004). Since we do not find any press coverage regarding the impending removal of step-up until January 2005 (see Appendix F), we take 2005 to be the relevant announcement year when considering anticipatory responses in our empirical analysis. The removal of step-up for stock meant that, from January 1, 2006, intergenerational transfers of appreciated stock (including unlisted stock and equity-based mutual funds) received carryover basis treatment.[22] That is, heirs who received appreciated stock continued to be subject to inheritance tax on the value of the stock

[19] The capital gains tax rate was 28% from 1992 to 2013 and has never depended on individual income (i.e., there is no progressive rate structure). The same rate was also applied to other forms of capital income, such as interest income. Net capital losses are fully deductible against ordinary income at the capital gains tax rate.

[20] Norway's inheritance tax had a rate structure that depended on the cumulative amount of the inheritance received, and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction was made between *inter vivos* gifts and bequests at the time of death. Appendix Figure A1 and Table A2 provide details on the rates and brackets over time.

[21] Further detail on later policy changes, including the removal of step-up for non-stock asset classes (concurrent with the removal of the inheritance tax in 2014), can be found in Appendix F.

[22] In Norway, mutual fund investors are not taxed when fund managers buy and sell stocks or reinvest dividends. Tax is only triggered when the investor himself sells his mutual fund units or when the investor receives taxable distributions.

they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period.[23] Losses, however, were not permitted to be carried over in the 2006 reform, so stocks with losses continued to be subject to a "step-down" in basis at the time of transfer. Step-up also continued for non-stock assets (such as real estate) until 2014.[24]

Our empirical analysis focuses on how individuals reacted to the 2005 announcement of step-up's impending removal, rather than the later implementation of that removal. The announcement date is important because the capital gains tax treatment of the original owner is not changing. The original owner faces the same cost of realizing his capital gains before or after step-up is removed. But the announcement of step-up's removal informs the original owner that the inheritance he may leave in the future will be more expensive, which will cause him to reassess his trade-off between present realizations and future intergenerational transfers. We formalize this understanding of step-up's effects on owners of unrealized gains in the next section of the paper.

Although repealing stepped-up basis and inheritance taxes are frequently considered in tandem, our empirical setting allows us to isolate the effects of removing step-up from any changes to the inheritance tax regime. Conceptually, an inheritance tax is an imperfect substitute for exempting inherited capital gains from the capital gains tax base. Inheritance transfer taxes apply to the current value of the bequest, regardless of any unrealized capital gains. Under stepped-up basis, an investor who sells a highly appreciated asset the day before death and leaves the resulting cash as inheritance is subject to both the capital gains tax and the inheritance tax, whereas a dying investor who bequests the unrealized asset itself pays only the inheritance tax. As our conceptual framework in Section 4 demonstrates, three distinct policy levers affect the trade-off between capital gains realizations and intergenerational transfers: the capital gains tax rate, the inheritance tax rate, and step-up.[25]

[23] From 2006 until the removal of the inheritance tax in 2014, recipients of gifted or inherited stock had their inheritance tax values reduced by an amount equal to 20% of the carried-over capital gains, to lessen the "double taxation" from being subject to both an inheritance tax and carried over capital gains. We abstract from this deduction in our explanation of the policy changes, but incorporate it explicitly where magnitudes matter, such as in the MVPF calculation of Appendix C.

[24] See Appendix F for additional detail on both the 2006 reform and the subsequent removal of step-up for non-stock assets, which is not the focus of our empirical analysis.

[25] There is a small existing literature on the interactions between inheritance and capital gains taxation. Prior work illustrates two of the three "policy levers" mentioned above: Auten and Joulfaian (2001) show that higher estate tax rates encourage more capital gains realizations by the original owner. Poterba (2001) provides evidence that larger unrealized capital gains push owners towards the utilization of stepped-up basis. We are the first paper to show empirically that changing stepped-up basis affects taxable realizations.

3.1 Other Elements of the 2006 Tax Reform

The repeal of step-up was concurrent to a broader 2006 reform of shareholder taxation, which generated a large (anticipatory) increase in dividend payouts at closely-held firms before 2006 and a sharp decline afterward (Alstadsæter and Fjærli, 2009). If owners of closely-held firms substitute from dividends to capital gains in response to these other changes, this could complicate our attribution of the increase in capital gains realizations in Section 5 to the removal of stepped-up basis.[26] However, we do not think any such substitution is likely to drive results. First, the timing does not align: our results in Section 5 show an increase in realized capital gains starting in the announcement year of 2005, when dividend payouts were high (and still tax-advantaged), and our results are stable in later years as dividend payouts fell.[27] Secondly, we do not think closely-held firms were likely to engage in dividend-to-capital gains substitution, both ex-ante (because closely-held firms are illiquid and capital gains were not tax-advantaged relative to dividends) and ex-post (because our results show that the capital gains increase we document is concentrated in listed equity & mutual funds). Finally, shareholder deductions accumulated under the old system remained usable for realizations after 2006 – only the accumulation of future deduction amounts changed (The Norwegian Tax Administration, n.d.). Therefore, there was no discrete change to the tax calculation of realizing capital gains in 2005 vs. early 2006; the same deduction amounts would be available both times. Additionally, prior literature has documented that both the pre- and post-reform systems were designed to treat dividends and capital gains broadly symmetrically (Sørensen, 2005; Alstadsæter and Fjærli, 2009), so the reform did not create an obvious first-order incentive to substitute taxable realizations for dividends.

4 Conceptual Framework

This section presents a lifecycle model with taxes on capital gains realizations and bequests. Our model follows a broad literature on lifecycle portfolio models (surveyed in Gomes (2020)) and is more specifically an extension of the model in Dammon et al. (2001). We use this model as a framework to understand the effects of removing step-up. The removal of step-up reduces the after-tax value

[26] A competing explanation would need to account for increased capital gains realizations concentrated among older individuals and those with higher baseline portfolio appreciation.

[27] It is also important to note that our realization results look at the outcome of taxable realized capital gains (a component of individual taxable income), which is net of all deductions and does not include any tax-free company transformations (i.e., the movement of shares into personal holding companies) occurring during this time period. We track basis across tax-free reorganizations and look through up to 15 levels of firm ownership to form our estimates of unrealized capital gains, so our results are not affected by whether operating companies are held directly or through holding companies.

of an inheritance in the hands of the heir. Under the assumption that givers of intergenerational transfers have altruistic bequest motives (i.e., that they care about the taxes their heirs pay on the inheritance), we show that substitution effects from this price change will encourage capital gains realizations during the original owners' lifetimes. We also simulate the model (it is not solvable analytically) in order to discuss dynamics and heterogeneity in the expected response.

4.1 Lifecycle Problem

Consider an individual who may live from time $t = 0, \dots T$, where $\mathcal{T} \in \{1, ..., T\}$ denotes the stochastic period of death. Death occurs by period T with certainty and mortality risk is exogenous and independent of asset price shocks. The individual's wealth at the beginning of period t is W_t. This wealth is partially held in the form of a risky asset: the individual owns n_{t-1} shares priced at P_t at the beginning of time t. These shares contain an amount of unrealized capital gains U_t at the beginning of period t. The individual holds the rest of his wealth in a risk-free asset earning an after-tax nominal return of r. In each period $t < \mathcal{T}$, the individual chooses after-tax consumption (c_t) and shares in the risky asset to hold into next period (n_t). The individual faces capital gains tax (τ_c) on realized gains when he sells risky assets. The price of the risky asset changes between periods and wealth and unrealized capital gains evolve accordingly. In period $\mathcal{T}$, the individual dies and remaining wealth passes to his heir as a bequest (with post-tax value $B_{\mathcal{T}}$). Heirs face inheritance tax (τ_I) on the value of any bequest. When there is a carryover basis regime (denoted with the policy indicator $\theta = 1$), the heir will also have to pay capital gains tax (τ_c) to realize their predecessor's terminal unrealized capital gains.

The individual maximizes utility subject to per-period budget constraints and a function that transforms wealth at death into a post-tax bequest amount:[28]

$$\begin{aligned}
&\max_{\{c_t, n_t \geq 0\}_{t=0}^{T-1}} \quad \mathbb{E}_0 \left[\sum_{t=0}^{\mathcal{T}-1} \beta^t u(c_t) + \beta^{\mathcal{T}} \nu(B_{\mathcal{T}}) \right] && s.t. \\
&W_{t+1} = \big(1 + r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1} && \forall t = 0, ...T-1; \\
&B_t = W_t\big(1 - \tau_I\big) - \theta \tau_c U_t^+ && t = \mathcal{T};
\end{aligned} \tag{1}$$

where r is the nominal return on the risk-free asset and $U_t^+ \equiv \max\{U_t, 0\}$ (in keeping with the

[28]Note that our setup assumes the heirs realize their inheritance immediately and that the giver cares about his heir's post-tax bequest amount. To alternatively assume that the heir realizes at some known time after the death of the predecessor, we simply need to discount the $\theta \tau_c U_t^+$ term in B_t.

fact that the 2006 Norwegian policy reform carried over gains but not losses). The amount of capital gains realized at time t, R_t, depends on whether there is an embedded gain or loss in the beginning-of-period shares and whether the investor chooses to buy or sell shares this period:

$$R_t \equiv \begin{cases} \max(1 - \frac{n_t}{n_{t-1}}, 0)U_t & \text{if } U_t > 0 \quad \textit{(embedded gain)} \\ U_t & \text{if } U_t \leq 0 \quad \textit{(wash sale on embedded loss)} \end{cases} \tag{2}$$

We assume that all shares will be sold and immediately repurchased whenever there is an embedded loss ($U_t < 0$).[29] Under an average basis rule, unrealized capital gains evolve according to:[30]

$$U_{t+1} = (P_{t+1} - P_t)n_t + \max(U_t, 0)\frac{n_t}{\max(n_t, n_{t-1})} \tag{3}$$

where the first term represents new appreciation from price changes in period $t+1$ and the second term represents accumulated appreciation from prior periods (if any). For notational convenience in future sections, we also define $\gamma_t^+ \equiv U_t^+/P_t n_{t-1}$ as the individual's positive unrealized capital gain expressed as a share of beginning-of-period stock wealth.

4.2 Realizations under Step-Up versus Carryover

We now use the model to characterize how the tax treatment of inherited capital gains affects realization choices during the original owner's lifetime. Under step-up ($\theta = 0$), the heir's basis is reset and the predecessor's terminal embedded gains do not generate a tax liability for the heirs. Under carryover ($\theta = 1$), the heir inherits the predecessor's basis and therefore inherits latent capital gains tax liability on positive terminal gains U_T^+.

[29] Without transaction costs or wash sale restrictions, investors with losses will optimally liquidate and then immediately repurchase all shares. Since realizations in the model arise only from net reductions in shareholdings, sale-and-repurchase of gain positions is not in the choice set.

[30] Under this formula, the tax basis for currently held shares is the weighted average purchase price of those shares. This is an average basis assumption and is necessary for tractability in models with more than a few periods since a (more realistic) first-in-first-out basis assumption requires separate state variables for each asset purchase (Dybvig and Koo, 1996). DeMiguel and Uppal (2005) show, in a model broadly similar to our own, that the certainty equivalent loss from using the average tax basis instead of the exact basis is small. Our empirical results are also robust to calculating unrealized capital gains under either basis assumption (Appendix Figure A8).

4.2.1 Baseline Prediction: Increased Realizations under Carryover

We can rewrite the giver's problem for state variables $X_t = [P_t, U_t, n_{t-1}, W_t]'$ as

$$V_t(X_t;\theta) = \mathbb{P}(\mathcal{T} > t | \mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \left\{ u\big(c_t\big) + \beta E_t[V_{t+1}(X_{t+1};\theta)] \right\} + \mathbb{P}(\mathcal{T} = t | \mathcal{T} \geq t)\nu\big(B_t\big) \quad s.t.$$

$$W_{t+1} = \big(1+r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1}$$

$$B_t = W_t(1-\tau_I) - \theta \tau_c U_t^+$$

It is clear that the removal of step-up ($\Delta\theta$) affects the giver only through the payoff at death, $\nu(B_t)$. The change in this death payoff from a marginal policy move from step-up towards carryover is:

$$\frac{\partial \nu(B_t)}{\partial \theta} = -\tau_c U_t^+ \cdot \nu'\big(W_t(1-\tau_I) - \theta\tau_c U_t^+\big) \leq 0$$

This is the utility cost of the policy change to the giver if he dies in period t. Since ν is increasing (givers value bequests), removing step-up hurts givers with $U_t > 0$ and has no effect when $U_t \leq 0$. The switch to carryover reduces the post-tax value of bequests relative to lifetime realizations for the giver. If substitution effects dominate income effects along this margin, we expect the giver to choose a weakly higher level of taxable realizations under carryover than under step-up.[31]

4.2.2 Heterogeneity in Exposure (Comparative Statics)

The following discussion characterizes how exposure to (i.e., current period utility costs from) the removal of step-up differs across investors. If substitution effects dominate income effects, these exposure gradients imply heterogeneous treatment effects of the reform on lifetime realizations, which we exploit in our empirical strategies in Section 5.

Remark 1: Exposure by Age. Conditional on his subjective distributions over (W_s, U_s) at possible death dates s, the closer an investor is to his time of death the more the removal of step-up matters for his current choices. Formally, for an investor alive at the beginning of period t:

$$\frac{\partial V_t}{\partial \theta} = \sum_{s=t}^{T} \beta^{s-t} \cdot \mathbb{P}(\mathcal{T} = s \mid \mathcal{T} \geq t) \cdot \mathbb{E}_t \left[\frac{\partial \nu(B_s)}{\partial \theta} \right].$$

since θ only enters the death payoffs of the Bellman equation. The weights $\beta^{s-t}\mathbb{P}(\mathcal{T} = s | \mathcal{T} \geq t)$

[31] In order to translate these price changes into behavioral predictions, we need to assume that substitution effects dominate income effects along the realization-bequest margin and that the magnitude of the substitution effect is monotone in the size of the price change. These assumptions will be implicitly maintained by the parameterization of the model in our numerical simulations.

show the age channel: holding the expected death-state derivative fixed across remaining horizons, moving expected mortality sooner (as happens when age rises) makes the policy's effect less heavily discounted.[32] All else equal, older investors are therefore more exposed to the removal of step-up. In our analysis of the removal of step-up for stock in Norway in Section 5, this observation guides our primary age-based empirical strategy.

Remark 2: Exposure by Unrealized Gains at Death. Holding age and beliefs about W_s fixed at each possible death date s, investors who expect to die with more taxable unrealized capital gains are more exposed to the removal of step-up. Differentiating the death payoff at $\theta = 0$,

$$\frac{\partial \nu(B_s)}{\partial \theta}\Big|_{\theta=0} = -\tau_c U_s^+ \cdot \nu'(W_s(1-\tau_I))$$

shows that the welfare cost is weakly increasing in U_s at each W_s (strictly so on gains). Formally, a first-order stochastic increase in $U_s \mid W_s$ at each possible death date therefore raises the magnitude of $\partial V_t/\partial\theta$. Investors who expect to die with more embedded gains at a given terminal wealth level lose more from the removal of step-up and therefore (conditional on age and wealth) have a stronger substitution effect towards lifetime realizations. This observation guides our second empirical strategy exploiting heterogeneity across portfolio positions.

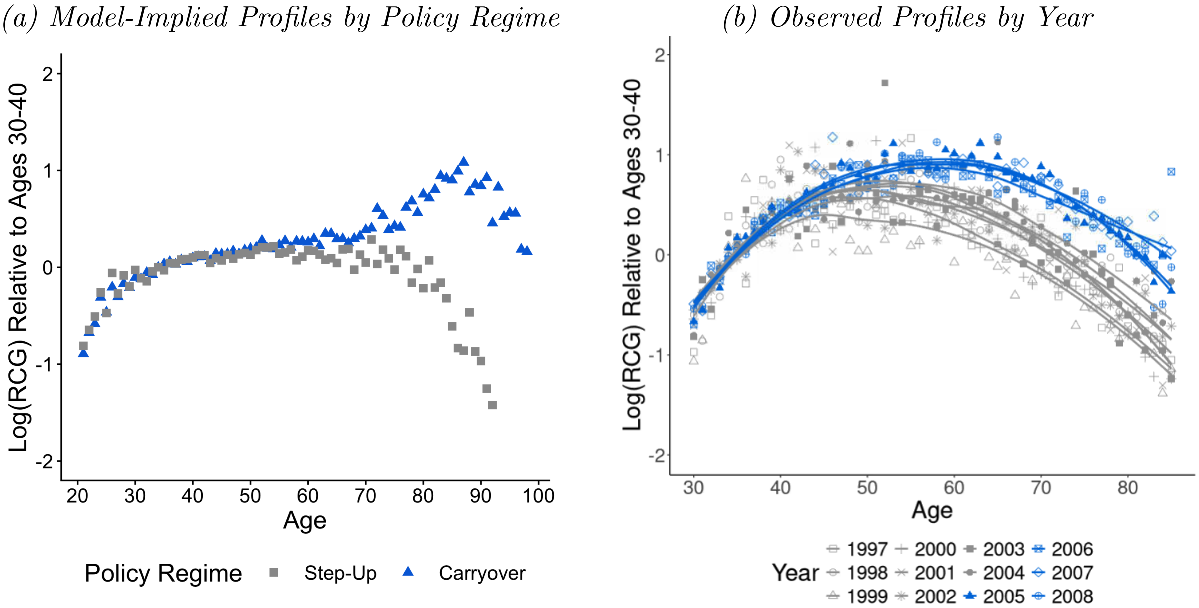
4.3 Lifecycle Realizations under Step-Up and Carryover

Thus far, we have focused on impact exposure to the removal of step-up: we have examined comparative statics holding investor portfolios fixed. Our empirical analysis, however, will exploit a change in policy which will induce both short-term (transitional) and long-term (equilibrium) responses to the removal of step-up. Our model simulations here compare realizations when one or another of the two policy regimes is always in place, deferring further discussion of the transition dynamics between them to the empirical section. For these simulations, we parameterize the model with $T = 80$, mortality following Norwegian life tables, isoelastic utility (implying choices scale with wealth) and an exogenous binomial process for the price evolution of the risky asset. Further details and parameters for the model simulation are described in detail in Appendix B.

Both the level and shape of capital gains realizations over the lifecycle differ between the step-up and carryover policy regimes. In the step-up regime, individuals hold a larger share of their

[32] Age also shifts the death-payoff terms $\mathbb{E}_t[\partial\nu(B_s)/\partial\theta]$ themselves, since investors who die later have had more time to accumulate wealth and embedded gains; Remark 1 isolates the mortality-timing channel by holding these fixed across horizons. The simulations solve the full model and capture both channels.

Figure 2: Realized Capital Gains by Age, by Policy Regime



Note: This exhibit displays two plots of the profile of log realized capital gains in age. Y-axis values are normalized to the average value for the 30-to-40 year-olds in the same year or simulation. Panel (a) plots age-log(RCG) profiles from our model simulation when either step-up or carryover is always in place (i.e., when the policy regime does not change during the simulated individuals' lifecycles). Parameterization details are described in Appendix B. Relative to the step-up regime, older individuals in the carryover regime realized more capital gains. Panel (b) plots the empirical analogue: logged average realized capital gains by age in our data from 1997-2008. The points are year- and age-specific logged averages. The smoothed lines depict local nonlinear regressions (LOESS) of logged average realizations on age for each year or era. Years prior to 2005 (during step-up) are depicted in light grey and the years from 2005 to 2008 are depicted in dark blue. The shape of the profile is relatively stable from 1997 to 2004 (during step-up), but older individuals realized relatively more since 2005. We end this graph in 2008, before the 2009 decrease in inheritance tax rates.

portfolio in equity throughout the lifecycle and decrease capital gains realizations late in life. In the carryover regime, equity is drawn down as individuals age, implying more taxable realizations late in life (Figure 2 Panel (a)). Therefore, moving from a step-up regime to a carryover regime entails two types of realization responses: (i) in the short term, people who built up equity shares that are higher than optimal for the carryover regime need to reduce their equity holdings, which requires taxable sales and (ii) the carryover regime has persistently higher levels of realizations. Both of these effects are stronger for older individuals. The short-term effects are increasing in age because older people have a larger gap between optimal equity holdings under the two policy regimes (and thus a larger need to sell). The long-term effects are increasing in age because older people in the carryover regime realize more. In the next section, we begin our empirical analysis of the removal of step-up, guided by the framework and heterogeneity discussed above.

5 Observed Responses to the Removal of Step-Up

In 2005, following thirteen years without substantial changes to capital gains and inheritance tax rates, Norway announced that step-up would be removed for inherited capital gains from stock (including private business interests and equity-based mutual funds).[33] Stock gifted or bequeathed after January 1, 2006, would be subject to capital gains tax on the original owner's gains when the heir next realized the asset.

In this section, we document responses to the 2005 announcement of the removal of step-up for stock.[34] Aggregate capital gains realizations more than tripled between 2004 and 2005 (Appendix Figure A2), but year-to-year variation in aggregate realizations is frequently large and causal attribution requires a research design. We first estimate average treatment effects on the treated using a difference-in-differences design that exploits the model-implied age gradient in exposure (Section 5.1). We then turn to assessing the treatment margin. Model Remark 2 predicts that the response should scale with embedded appreciation. Our second design tests this prediction directly, exploiting heterogeneity in pre-2005 stock portfolio appreciation among stockholders of the same age and wealth (Section 5.2). The two designs produce similar aggregate results through the end of our data in 2018.

5.1 Age-Based Design

Our primary empirical strategy builds on two intuitions. First (following Remark 1 from our model), we expect realization responses to the removal of step-up to be increasing in age, with younger age groups being effectively "untreated" by the removal of step-up. Second, we think the removal of step-up is relatively unusual in the way it affects the age-realizations gradient: we expect that normal year-to-year macroeconomic fluctuations would affect the realization behavior of individuals of different ages more proportionally.

These intuitions lead us to a continuous differences-in-differences design (formalized below) in which treatment exposure is increasing in current age. Our primary specifications collapse the continuous age gradient in exposure to a binary difference-in-difference design comparing groups of older and younger individuals. This binarization allows us to identify summary average treatment

[33]See Appendix Table A2 for detail on minor modifications to the inheritance tax brackets during this time.

[34]Although step-up was eventually removed for non-stock asset classes in 2014 (see Appendix F), evaluation of that removal is confounded by the 2014 removal of the inheritance tax, which had a simultaneous but opposite effect on the relative price of bequests versus realizations. There was also a capital gains tax cut in 2014. We therefore choose to focus our analysis only on the earlier reform, which removed step-up for stock while holding inheritance and capital gains tax rates fixed.

effects for the group of older individuals (Callaway et al., forthcoming) and to cleanly map those effects into an overall impact on capital gains tax revenue in Norway. However, we also examine heterogeneous treatment effects by age and find them to be consistent with our theoretical predictions, which we interpret as bolstering the case for our causal claims.[35]

5.1.1 Identifying Assumptions

To identify our treatment effects, we will assume that if step-up had not been removed, different age groups' average realizations would experience the same percent changes over time in response to macroeconomic shocks and other time effects. Let $A_{i,t}$ denote a normalized age variable with support $\mathcal{A}$, normalizing $A_{i,t} = 0$ as the control group of "young people" who are unaffected by treatment at time t. Our assumptions condition on X_i, a vector of time-invariant characteristics with support $\mathcal{X}$. (In practice, X_i denotes the individual's position in the age-specific wealth distribution and an indicator for private business ownership). We assume:

Assumption 1 (Parallel Trends in Percent Changes)**.** *The average realizations of individuals with age a would have, in the absence of treatment, evolved proportionally to the average realizations of the untreated younger age group ($A_{i,t} = 0$):*[36]

$$\frac{E[Y_{i,t}^0|A_{i,t} = a, X_i] - E[Y_{i,t_0}^0|A_{i,t_0} = a, X_i]}{E[Y_{i,t_0}^0|A_{i,t_0} = a, X_i]} = \frac{E[Y_{i,t}^0|A_{i,t} = 0, X_i] - E[Y_{i,t_0}^0|A_{i,t_0} = 0, X_i]}{E[Y_{i,t_0}^0|A_{i,t_0} = 0, X_i]} \quad \forall t, a, x$$

where $Y_{i,t}^0$ is individual i's potential outcome (realized capital gains) in year t in the absence of treatment and $t_0 < 2005$ is a fixed pre-treatment base year. This parallel trends assumption is equivalent to assuming that, in the absence of the policy change, yearly shocks may move the age-log(realizations) profile up and down, but the curvature of the profile (i.e., the relative realization levels of different ages) will be stable over time. We investigate how the shape of the observed age-log(realizations) profile changed over time in Figure 2 Panel (b), comparing each age's average realizations to the average realizations of 30-to-40 year-olds in the same year. In the pre-reform period (before 2005), the shape of this profile appears to be roughly stable across years. Averaging

[35] We do not interpret differences between age groups as an estimate of the average causal response to marginal exposure, which would require additional selection restrictions to identify (Callaway et al., forthcoming).

[36] An assumption of parallel trends in percent changes is commonly made by assuming parallel trends in a logged variable — but zeros in our outcome variable make a log transformation particularly inappropriate to our setting (Chen and Roth, 2024). Furthermore, estimating a diff-in-diff using OLS on a log transformation of the dependent variable only approximates the proportional difference in growth rates between treated and untreated groups. Such approximations ignore Jensen's inequality and thus may deliver biased estimates of the true multiplicative effect (Ciani and Fisher, 2019). Our estimator avoids this bias and can be estimated with Poisson Pseudo Maximum Likelihood (PPML) regression. PPML consistently estimates the multiplicative effect as long as the mean function is correctly specified as multiplicative (Wooldridge, 2023).

across pre-period years, realizations peaked around age 54 at a level approximately 94% higher than those of 30-to-40 year-olds, and then decreased for older ages, so that average realizations at age 70 were approximately 5% below the realizations of 30-to-40 year-olds. This inverted-U relationship is consistent with the model-implied profile from our model in Figure 2 Panel (a). Figure 2 Panel (b) also provides initial evidence that this relationship changes in the initial years after the announcement of step-up's removal in 2005: realizations peaked at age 60 at a level approximately 160% higher than those of 30-to-40 year-olds and declined slowly, so that realizations by 70-year-olds were 119% higher than those of 30-to-40 year-olds in 2005-2008.

We also require a "no anticipation" assumption for the years we consider to be our pre-period:

Assumption 2 (No Anticipation Before 2005)**.**

$$E[Y^0_{i,t}|A_{i,t} = a, X_i = x] = E[Y^a_{i,t}|A_{i,t} = a, X_i = x] \quad \forall t < 2005,\ a \in \mathcal{A},\ x \in \mathcal{X}$$

where $Y^a_{i,t}$ is individual i's potential outcome (realized capital gains) in year t if he experienced an exposure level like that of age a. Conditioning on X_i is only meaningful if untreated individuals are observed at every covariate value attained by the treated, so we additionally require:

Assumption 3 (Overlap)**.**

$$P(A_{i,t} = 0 \mid X_i = x) > 0 \quad \forall x \in \mathcal{X}$$

Under Assumptions 1 – 3, we can employ a difference-in-difference estimator to identify the effect of the removal of step-up in terms of average percent changes from the counterfactual and group-specific average treatment effects $ATT_t(a) = E[Y^a_{i,t} - Y^0_{i,t}|A_{i,t} = a]$ for each age a.[37] We can also identify a binary summary effect for all "old" versus "young" individuals: $ATT_t = E[ATT_t(a)|A_{i,t} > 0]$.[38] We are interested in the ATT_t in levels because it translates directly into the aggregate increase in capital gains tax revenue.

[37]Our $ATT(a)$ measures the average difference in potential outcomes between receiving dose a and dose 0 for the age group that was actually treated with dose a.

[38]Callaway et al. (forthcoming) refer to this summary parameter (in the absence of additional restrictions on heterogeneous treatment effects) as ATT^{loc}. It is "local" in the sense that it is built from each age group's effect at the dose that age group actually received, rather than from the effect that dose a would have had on the entire treated population. Given our focus on the policy impact of the removal of step-up, this parameter is the right one for estimating the causal increase in capital gains revenue from Norway's actual policy change.

5.1.2 Control & Treatment Group Definition

Although our model implies that younger age groups will have less exposure, the model does not tell us what age cutoff we should use to form the control group of plausibly-unexposed younger individuals. To interpret our treatment effects as exact estimates, we need to believe that the younger ages we pick to form our control group are truly untreated by the removal of step-up. If the younger age group is partially treated and increases realizations in response to the removal of step up, the treatment effects we estimate for the older groups will understate the true treatment effects, since the comparison will net out any realization increase from the younger group. In practice, we set up a difference-in-difference empirical strategy using 30-49 year olds as our control group and test our estimates' robustness to the choice of other age cutoffs in Appendix Figure A4. We accept the fact that our estimates may be lower bounds if 30-49 year-olds are partially treated.

5.1.3 Estimation

Our primary outcome is net realized capital gains from all taxable asset categories as reported on the individual's annual tax return.[39] We first estimate a binarized regression that compares individuals aged 30-49 ($A_{i,t} = 0$) to individuals aged 50+ ($A_{i,t} > 0$):

$$Y_{i,t} = exp\Bigg(\beta_0 + (\mathbb{1}[A_{i,t} > 0] \times Post_t)\beta_1 + \mathbb{1}[A_{i,t} > 0]\beta_2 + Post_t\beta_3 + X_i'\beta_4\Bigg)\epsilon_{i,t} \tag{4}$$

where X_i are time-invariant control variables: a saturated set of indicators for the individual's position in the age-specific wealth distribution and for private business (unlisted stock) ownership. $Post_t$ is an indicator that equals one if the year is greater than or equal to 2005 (the announcement year); and $E[\epsilon_{i,t}|A_{i,t}, Post_t, X_i] = 1$. We also estimate event study estimates by replacing the $Post_t$ variable in the equation above with an indicator variable for each specific year. The exponentiated coefficient $\exp(\beta_1)-1$ gives the percent change in realizations attributable to the reform, and we can construct the summary average treatment effect in levels (the ATT) by averaging the difference between each treated observation's post-period predicted value and its counterfactual predicted

[39] The tax data nets out gains and losses across all realizations within an individual and year. We set net losses to zero in our regressions, as PPML requires a non-negative outcome and the 2006 reform does not incentivize changes in loss-realizing behavior, which, given full deductibility against ordinary income, should optimally occur whenever losses accrue.

value setting $\beta_1 = 0$. To examine how responses vary by age, we run the following regression:

$$Y_{i,t} = exp\Bigg(\beta_0 + \sum_{a \in \mathcal{A}^+} (\mathbb{1}[A_{i,t} = a] \times Post_t)\beta_{1,a} + \sum_{a \in \mathcal{A}^+} \mathbb{1}[A_{i,t} = a]\beta_{2,a} + Post_t\beta_3 + X_i'\beta_4\Bigg)\epsilon_{i,t} \quad (5)$$

with $E[\epsilon_{i,t}|A_{i,t}, Post_t, X_i] = 1$. We recover the percent change for each age group $A_{i,t} = a$ and corresponding $ATT(a)$ from this regression.[40] All regressions are estimated with Poisson Pseudo Maximum Likelihood (PPML) and include all income tax filers aged 30+.

5.1.4 Threats to Identification in the Age-Based Design

Assumption 1 would be undermined if other policy or market shocks during our time period affect realizations in a way that has differential age effects. Our most concrete concern is with confounding policy shocks: Norway cut top inheritance tax rates in half in 2009 and then fully repealed the inheritance tax in 2014. Older individuals are differentially exposed to these cuts, and the cuts push against the realization response to the removal of step-up; we therefore believe our age design to be potentially downward-biased from 2009 onward. Two other potentially age-graded policies are discussed elsewhere: the concurrent 2006 reform of shareholder taxation, whose incidence fell on the (likely older) owners of closely-held firms (Section 3.1), and the possibility that our 30–49 year-old control group is itself partially treated (Section 5.1). Market shocks could plausibly also have differential effects by age: older investors have held their positions longer and carry more embedded appreciation, so the 2006–2007 run-up and the 2008 crash could have moved their taxable gains differently even absent the reform. Our multiplicative specification absorbs market shocks that raise stock turnover by the same proportion across portfolios, but not shocks that affect which mix of positions are sold. Because our pre-period contains a full boom-and-bust cycle (1999–2002) and the first two years of the subsequent boom, our event study pretrends can help evaluate this concern to the extent that it is common across boom-and-bust cycles. A final caveat about our age design is that we define treatment and control groups by contemporaneous age. Therefore the age profile is stable over time only if realization propensities do not differ across birth cohorts conditional on our controls.

[40] The choice of age bin is arbitrary, so we present results using both decade age bins (e.g., 50-59, 60-69, etc) and deciles of the observed age distribution.

Table 2: Realization Responses from Baseline (Age-based) Empirical Design

Difference-in-Difference Regressions (2000-2008)

	Specification:		*Implied Treatment Effects:*	
Ages in Treatment Group:	(1)	(2)	%Δ	ATT
All Aged 50+		0.277**	32%	$314
		(0.085)		
Ages 50-59	0.152		16%	$215
	(0.095)			
Ages 60-69	0.356***		43%	$452
	(0.090)			
Ages 70-79	0.504***		66%	$410
	(0.106)			
Ages 80-89	0.480***		62%	$233
	(0.106)			
Number of Treated Individuals	1,472,799	1,503,089		
Number of Total Observations	24,807,216	25,071,648		

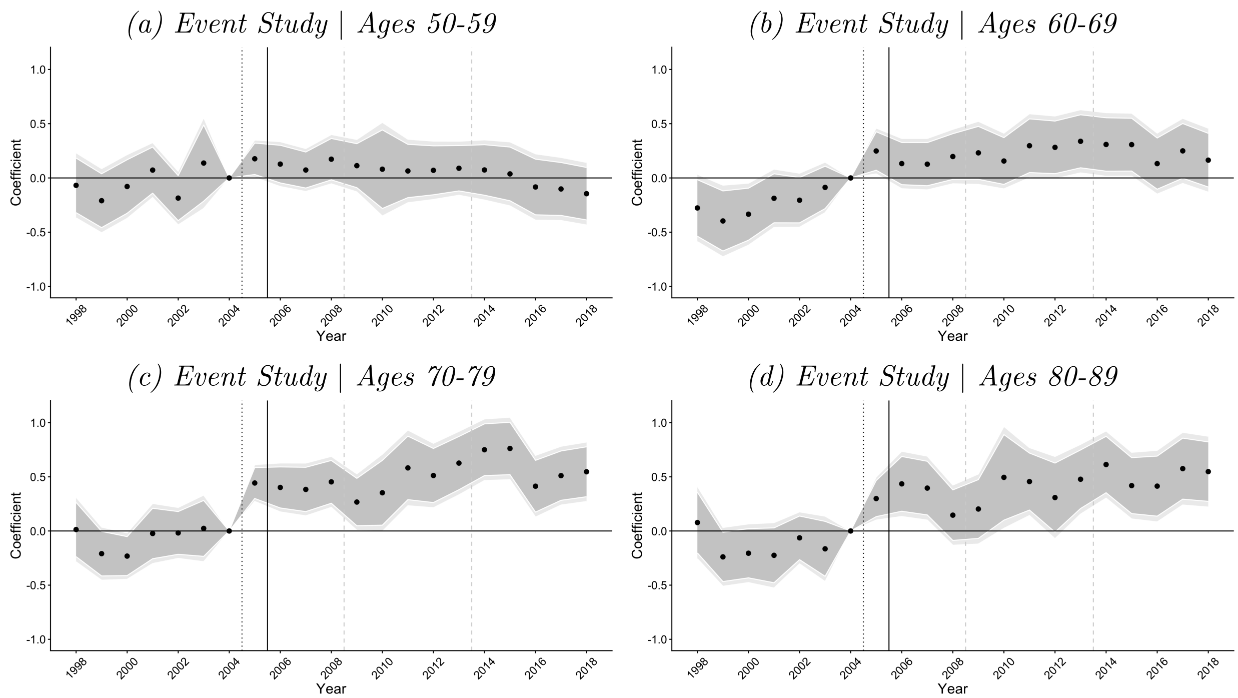
Note: This table presents estimates from our baseline empirical strategy which compares older age groups to a control group of individuals aged 30-49. We estimate Equations 4 and 5 using Poisson Pseudo Maximum Likelihood. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. An observation is an individual-year income tax filing and we include all income tax filers aged 30 and above. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the indicated age group and equal 0 if the individual is in the 30-49 age group. Standard errors clustered by birth cohort are presented in parentheses (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). All regressions control for an indicator for private business ownership and a bin of highest-ever household net worth. Individuals in these regressions are classified as private business owners if, since the age of 25, anyone in their household owned unlisted stock between 1993 and 2018. Individuals are assigned their household's highest net wealth percentile over the same period of time. See Figure 3 for event-study estimates from this design.

Table 3: Realization Responses by Age Group & Type of Capital Gain (2000†-2008)

	Age Range of Treatment Group:				
ATT by Type of Capital Gains:	50-59	60-69	70-79	80-89	All 50+
All	215	452***	410***	233***	314**
Real Estate	-23	-42	-16	-22	-26
Private Business	-23	21	-38	58***	-5
Listed Stock, Mutual Funds, & Other	176*	272***	289***	219***	225***

Note: This table presents the implied treatment effects (ATTs) from re-running Equations 4 and 5 using different outcome variables corresponding to capital gains from particular asset classes. The first row ("All") reproduces the same output as in Table 2, using the outcome variable of net realized capital gains from any asset class, as reported yearly on each individual's tax return. Appendix Table A3 presents detailed regression tables underlying the other rows (other outcome variables). Sample, estimator and control variables are defined as in Table 2.
†: The years of data used in this table's estimates differ by row. Only our preferred outcome variable of "All" capital gains has full year coverage from 2000-2008. Capital gains for "Real Estate" are not observed in 2005 due to a data issue. Capital gains in "Private Business" and "Listed Stock, Mutual Funds + Other" are only observed from 2003 (when our stock transactions data begins). All analyses in this table end in 2008, before the 2009 decrease in inheritance tax rates. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Figure 3: Realization Responses from Baseline (Age-based) Empirical Design



Note: This figure depicts event study estimates corresponding to the difference-in-difference estimates presented in Table 2. Each event study compares the specified older age group to a control group of individuals aged 30-49. The light (dark) grey ribbon depicts 95% (90%) confidence intervals from standard errors clustered by birth cohort. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates the cut to statutory inheritance tax rates. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of changing capital gains tax rates.

5.1.5 Results from the Age Design

Table 2 presents estimated short-run responses from difference-in-difference regressions. Our summary measure implies that individuals aged 50 and above increased realized capital gains by 32%, or \$314 a year on average from 2005–2008. These effects are large compared to existing estimates of realization responses to capital gains tax cuts (see Appendix E). Effects are initially increasing in age, from $\beta_1 = 0.152$ for ages 50-59 (a percent change of 16%) to $\beta_1 = 0.504$ for ages 70-79 (a percent change of 66%). The estimated effect for individuals aged 80-89 is statistically indistinguishable from that of the 70-79 age group. These coefficients translate into level increases in taxable capital gains realizations ($ATT(a)$) ranging from \$215 for individuals aged 50-59 to \$452 for individuals aged 60-69.[41] Figure 4 Panel (a) depicts age-specific difference-in-difference coefficients for finer age groups (deciles of the observed over-50 age distribution). Effects are zero around age

[41] Note that, despite larger percent changes at older ages, the level response is largest for the 60-69 group due to higher baseline realizations.

50 but increase to a peak of $\beta_1 = 0.6$ (82%) around age 70, remaining fairly stable for the rest of the age distribution.

Table 3 explores what type of capital gains drive the increased realizations following the removal of step-up. For a subset of our data years, we are able to decompose our outcome variable into realized capital gains coming from real estate, from unlisted stock and from a residual category that is primarily listed stock and mutual funds.[42] Capital gains realizations from taxable real estate show small, statistically-insignificant negative effects, which serves as a useful placebo: the initial removal of step-up didn't apply to real estate. Capital gains realizations from unlisted stock reach conventional levels of statistical significance only for the oldest age group, where they account for roughly 25% of the total increase. For younger age categories, it is less clear whether there was a response in unlisted stock: results are small, statistically insignificant and sometimes negative. The response to the removal of step-up appears to be concentrated in the residual category dominated by listed stock and mutual funds, suggesting that the realization response came primarily from liquid asset sales rather than sales of private business interests. This is despite the fact that the majority of unrealized capital gains are in private businesses (Table 1).

Our estimates for 2009–2018 provide evidence that the differential realization response did not disappear over time, but these later estimates are exposed to additional (downward-biasing) policy changes and cannot be interpreted as equally clean causal effects. Figure 3 presents event study results from the age-based design which extends through 2018. Effects are clearly strongest for older ages. For the two oldest age groups, pre-trends estimates are flat on average (with some non-monotonic fluctuations), the realization response begins sharply in 2005, and the yearly coefficient estimates remain elevated through the end of our window with minimal decay toward the pre-period level. The stability of our estimates through 2018 despite expected effects from short-term rebalancing (see Section 5.3) and the downward bias we expect from the post-2008 inheritance tax cuts is surprising. We return to this finding in Section 5.2, where the UCG-based design provides a complementary medium-run estimate. The absence of decay also bears, if only imperfectly, on the counterfactual timing of the induced sales. If the reform had primarily advanced realizations that would have otherwise occurred later in life, the elevated realizations of the early post-period should eventually be offset as missing sales in later years. We observe no such reversal through 2018, a pattern easier to reconcile with the response drawing down gains that would otherwise have been

[42] Our data on capital gains from specific asset classes does not fully cover the same years as our primary outcome measure of total capital gains. We explain these limitations in more detail in Section 1 and in the note to Table 3.

held toward death. We caution, however, that fourteen post-reform years cannot rule out reversal at longer horizons, particularly for the younger treated cohorts, and that the confounds discussed above make the later years of this window less cleanly interpretable.

We now revisit our identification concerns from Section 5.1.4 in light of our results. First, our event study coefficients during the boom years of our pre-period (1999-2000 and 2003–2004) are, if anything, opposite in sign to the positive coefficients on the post-reform boom years (2005–2006). Our results therefore seem unlikely to be driven by differential age responses to boom-and-bust cycles. Our negative & insignificant results for taxable real estate are reassuring since real estate was not affected by the step-up reform, but might plausibly be affected by other confounding policy or market shocks. Similarly, the concentration of our response in listed stock and mutual funds somewhat alleviates concern about dividend tax changes to closely-held businesses driving the response. Finally, the fact that our results are increasing in age is consistent with the predicted gradient from Remark 1. Figure 4 Panel (a) shows that the short-run (2005–2008) and medium-run (2009–2013) age gradient peaks are similar, in contrast to the shift over time we might expect if cohort rather than age effects were driving the response. In the next section, we explore an alternative empirical design in which we will estimate similar aggregate responses, despite different potential confounds to identification.

5.2 UCG-Based Design

The model generates cross-sectional exposure to the removal of step-up through two channels: age (Remark 1) and the amount of unrealized capital gains in terminal wealth (Remark 2). In this section, we develop a second empirical design to follow the insight of Remark 2: among individuals with the same age and expected terminal wealth, exposure should be increasing in the expected level of unrealized capital gains in that wealth, and there should be no exposure for those without embedded gains. Neither design is free of confounds. We lay out the UCG design below and, after presenting its results, compare results and interpretability issues to those of the age design in Section 5.2.4.

To take Remark 2 to the data, we would ideally compare the responses of individuals with different expected unrealized capital gains at death but the same age, wealth and purchase history. We will proxy for the model-relevant stock portfolio appreciation at death, $\gamma^{+}_{i,\mathcal{T}}$, with observed appreciation from just before the policy change, $\gamma^{+}_{i,2004}$. However, since UCG are mechanically determined by purchase prices and current values, there is no residual variation among people with the same current holdings and identical transaction histories. To disentangle exposure to the removal of step-up from exposure to inheritance tax changes (which depends on terminal wealth), we condition on age and current wealth, accepting that the residual variation is necessarily variation in transaction histories – either through different initial investments in the market (later equalized by variation in returns) or different rebalancing behavior after the initial investment.

5.2.1 Identifying Assumptions and Estimation

This design rests on the identification assumption that among individuals holding stock at the end of 2004, realization trends, but for the removal of step-up, would have had a stable gradient in γ^+_{2004} (the ratio of positive unrealized capital gains to stock wealth in 2004) over time. Formally, let $\gamma^+_{i,2004}$ denote individual i's ratio of unrealized capital gains in stock to total stock value at the end of 2004, with $\gamma^+_{i,2004} = 0$ for the control group. We require analogues of Assumptions 1–3, with $\gamma^+_{i,2004}$ replacing $A_{i,t}$ and with X_i expanded to include age bins and equity share alongside total wealth and private-business indicators. In words, we assume that in the absence of the reform, average realizations of investors with $\gamma^+_{i,2004} > 0$ would have experienced the same percent changes as those of the $\gamma^+_{i,2004} = 0$ group (investors with zero or negative end-of-2004 appreciation) with the same age, stock wealth and total wealth. Fixing $\gamma^+_{i,2004}$ at its end-of-2004 value ensures that treatment intensity is determined pre-announcement and is not endogenous to the realization response itself.

We restrict the sample to individuals holding stock at the end of 2004. As in Section 5.1, the outcome is again net realized capital gains as reported on the individual's annual tax return. We estimate the PPML specifications of Equations 4 and 5 with $\gamma^+_{i,2004}$-bin indicators in place of age-bin indicators, and an event study specification interacting treatment with year. Our binarized treatment group pools all investors with $\gamma^+_{i,2004} > 0$. Because this design requires conditioning on age and end-of-2004 equity share as well as wealth, the control vector X_i here contains saturated age and equity-share-bin indicators in addition to the wealth-position and private-business indicators of Section 5.1.

5.2.2 Threats to Identification in the UCG-Based Design

This design compares individuals within the same age and wealth group who have different unrealized capital gains ratios (γ^+_{2004}) at the end of 2004. Necessarily, high-gamma individuals will have either made smaller initial investments and earned more on those investments or will have rebalanced less than their low-gamma counterparts. Group definition is therefore selected on a mixture of return luck, differences in initial investments and past non-realization.[43] Persistent traits between these groups of selected individuals (such as a persistently higher willingness to realize) will be absorbed by group fixed effects in our design. However, correlation between the return histories that generate high and low γ^+_{2004} and those individuals' responses to time variation remains a threat to identification in this setting.

This threat comes in at least three forms. First, winner-heavy portfolios may respond differently to market shocks. Our multiplicative specification absorbs symmetric changes to portfolio turnover, but not shocks that tilt sales toward winners rather than losers, or vice-versa. Second, transitory return shocks may move investors across the boundary of our definition of treatment and control, after which mean reversion in returns would bias our estimates downwards. Third, selecting the treatment and control groups based on 2004 gains could create a transitory mechanical gap in realizations between the groups, which would bias estimates upwards in the years immediately after classification and then fade to zero as the groups' gains converge. Each of these threats has empirical implications we take up after presenting the results: we will discuss these issues further once we have documented our estimates' lack of pre-trends across the 2000–2002 boom-bust cycle, that effects are concentrated far from the treatment boundary, and the timing and persistence of the realization response.

5.2.3 Results from the UCG-based Design

Table 4 Panel (a) presents the estimated short-run (2005–2008) responses for the UCG-based design. Column (1) estimates a difference-in-difference regression on all individuals who owned stock at the end of 2004 with a binary treatment variable equal to one if the individuals' end-of-2004 stock portfolio contains positive unrealized capital gains ($\gamma^+_{2004} > 0$).[44] Individuals with positive pre-2005 appreciation increased their realized capital gains by 48% (\$1,661/year on average) relative to the control group. Figure A3 plots event study coefficients separately by bin of γ^+_{2004}. The lowest bin (0-25% appreciated) shows no sustained positive response, whereas the higher bins show positive post-reform effects. Effects increase with γ^+_{2004}, consistent with Remark 2. Columns (2) and (3) of Table 4 re-estimate the binary specification restricting the sample to individuals who were less than or, respectively, at least 50 years old in 2005 (birth cohorts 1955 and earlier). Restricting

[43] Alternative choices of conditioning sets (such as an attempt to isolate the variation coming from return luck by conditioning on initial investments) necessarily let wealth depend mechanically on treatment intensity, reopening the differential exposure to the inheritance tax cuts that this design exists to close.

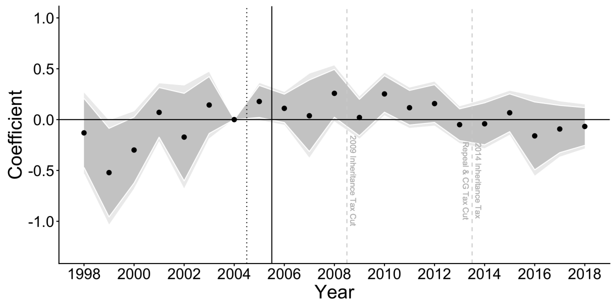
[44] Appendix Figure A4 shows robustness to redefining the treatment and control groups with alternative $\gamma^+_{2004} > 0$ cutoffs.

Table 4: Realization Responses from Alternative (UCG-based) Empirical Design

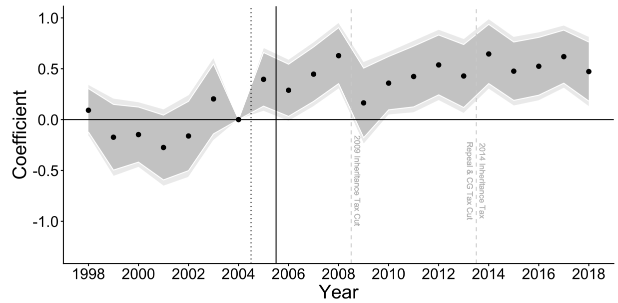
(a): Difference-in-Difference Regressions (2000-2008)

	Specification:		
	(1) All	(2) <50	(3) 50+
Indicator for $\gamma > 0$	0.390***	0.229*	0.508***
	(0.070)	(0.114)	(0.080)
Number of Treated Individuals	404,872	162,094	242,778
Number of Total Observations	4,489,058	1,820,266	2,668,792
Implied Treatment Effects			
Percent Change	48%	26%	66%
Average Increase (ATT)	$1,661	$1,044	$2,061

(b): Event Study | Aged <50 in 2005



(c): Event Study | Aged 50+ in 2005

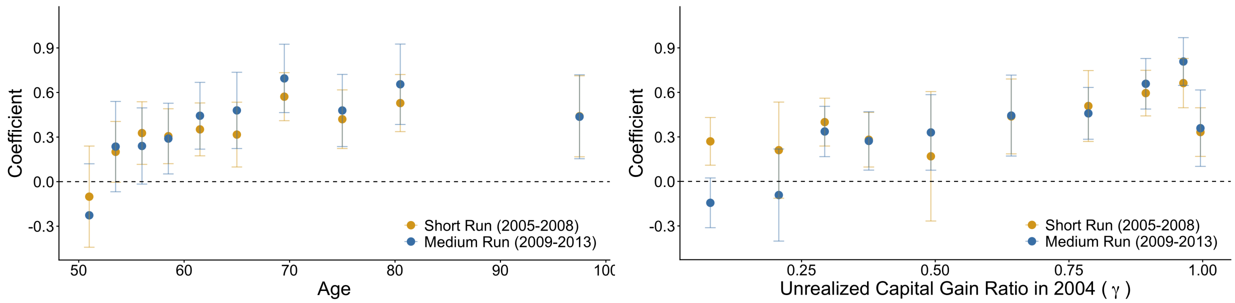


Note: This exhibit displays estimates from our alternative empirical strategy which categorizes individuals as treated if they have positive appreciation in their stock portfolio at the end of 2004. Panel (a) presents binary difference-in-difference estimates of Equation 4, estimated using Poisson Pseudo Maximum Likelihood. The first column ("All") displays estimates for all individuals aged 30 and above who owned stock at the end of 2004. To increase comparability with the age-based design, the next two columns re-run the regression on the sample of stockowners who were aged 30-49 ("<50") and aged 50 or above ("50+") in 2005. The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. All regressions include controls for age group, private business ownership, equity share bins, and age-specific rank in the household wealth distribution. Standard errors clustered by birth cohort are presented in parentheses (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.) To interpret our coefficients in terms of percent changes in the outcome variable, we need to exponentiate the coefficient and subtract one. Panels (b) and (c) display event study estimates broken out by age group (corresponding to the difference-in-difference estimates of columns 2 and 3 of Panel (a)). The light (dark) grey ribbon depicts 95% (90%) confidence intervals from standard errors clustered by birth cohort. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates the cut to statutory inheritance tax rates. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of changing capital gains tax rates. Appendix Figure A3 displays event study estimates broken out by treatment intensity (i.e., different levels of γ^{+}_{2004}).

to individuals who were at least 50 in 2005 serves two purposes. Substantively, it isolates the population for which the removal of step-up is most immediate. Practically, it also places the UCG design on (approximately) the same population as the age design, allowing the two designs' implied

Figure 4: Treatment Effect Heterogeneity, for Two Empirical Strategies

(a) Age-Based Design (b) UCG-based Design



Note: This figure shows short- and medium-run realization responses for our two empirical strategies by decile of treatment intensity. Our baseline age-based strategy compares older individuals (aged 50+) to a control group of individuals aged 30-49. The alternative UCG-based strategy compares individuals with more capital gain in their stock portfolio in 2004 to a control group of individuals with zero or negative capital gain in stock in 2004. All regressions control for age-adjusted wealth percentile and whether an individual owns a private business. The UCG-based regressions additionally control for age and equity share of the portfolio. Panels (a) and (b) show realization responses broken out by time period and the intensity of treatment (decile of age and decile of positive capital gains ratio, respectively). Treatment effects and 95% confidence intervals based on standard errors clustered at the cohort level are estimated for each decile of treatment in a diff-in-diff regression following Equation 5.

aggregate effects to be compared directly.[45] Panels (b) and (c) present event study estimates corresponding to the difference-in-difference regressions of Columns (2) and (3). For individuals under 50 in 2005, the event study has an upward pre-trend and post-period coefficients that are small and mostly statistically insignificant. The weakly positive binary estimate of Column (2) therefore reflects a pre-period baseline below the 2004 reference year rather than an elevated post-period. For individuals aged 50 and over in 2005, the event study shows smaller and statistically insignificant coefficients in the pre-period, after which the response begins sharply in 2005, and effects remain elevated through the end of our window with no decay toward the pre-period level.

The timing, persistence and structure of our estimates clarify some of the identification issues raised in Section 5.2.2. Regarding market cycles, event study coefficients do not seem to be correlated with the known boom-bust cycles of the pre- and post-periods. Appendix Figure A3 shows that treated portfolios near the boundary do display a pattern of mean reversion, but also that our results are driven by treated portfolios furthest from the boundary, where this is less of a concern. However, the shape of the under-50 event study (with a pre-trend rising from negative to zero and then post-period coefficients that fade out over time) is consistent with our concern about selection

[45]The populations are only approximately the same because the age design defines age groups based on contemporary ages, whereas column (3) of Table 4 subsets by age in 2005 (consistent with its treatment variable being fixed at end-of-year 2004). Over the 2005-2008 window, however, the two definitions nearly coincide: the year-t population aged 50+ differs from the 2005 cohort aged 50+ only through individuals aged 47-49 in 2005 who cross the threshold during the window.

into treatment causing a small transitory and mechanical response. Since investors under 50 face selection issues similar to older investors but experience the weakest incentive from the reform, their event study helps benchmark selection concerns. The response among investors aged 50+ is, in contrast, stable through 2018. Mechanical group convergence may still inflate its short-run level, and we read the short-term results of this design with that caveat. However, selection on end-of-2004 gain positions cannot produce the timing and persistence of our estimates, nor their model-consistent gradients in age and γ^{+}_{2004} (Figure 4).

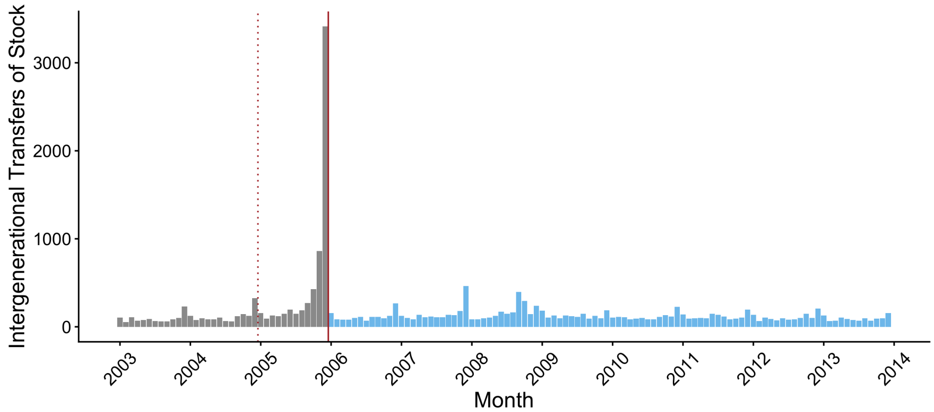
5.2.4 Results Comparison with the Age Design

While the age-based design runs on the full population, the UCG-based design subsets to the population of people who own stock at the end of 2004. This difference in populations means that the coefficients and level ATT estimates from the UCG-based and age-based designs are not directly comparable. The comparable objects are implied aggregates. Over 2005-2008, the age design implies an average annual increase in realizations of \$472 million; the UCG design restricted to individuals aged 50+ implies \$500 million per year. These estimates need not coincide, because each design nets out a different comparison group's response: the age design subtracts any response of individuals under 50, some of whom may hold embedded gains and face the reform's incentives through gifts and expected future bequests, while the UCG design subtracts the response of zero- and negative-gain holders.

The two designs share one potential confound and differ in the rest. Both compare investors who hold more embedded appreciation to investors who hold less, so both are exposed to market movements that affect winner-heavy portfolios differently; Sections 5.1.5 and 5.2.3 evaluate this concern for each design. Other confounds are distinct and, where their signs are known, opposite. The age design is biased downward from 2009 as the post-period inheritance tax cuts reduce older investors' realizations. The short-run results of the UCG design are biased upward as selection on end-of-2004 gain positions mechanically raises treated realizations in the years immediately after classification. The overall accordance between all our estimates across designs and across time periods suggests neither set of biases is large.

We take the age design results as our headline. Its identifying assumption does not select on the outcome's own history, its known biases are signed, and the 2005–2008 window in which we form our headline estimate precedes the inheritance tax cuts. We read the UCG design as establishing where the response is concentrated (in portfolios with high ratios of embedded gains), and as a

Figure 5: Anticipatory Inheritance Transfers in Late 2005



Note: This figure depicts the count of intergenerational transfers occurring by month in our stock transactions data. An outlier number of intergenerational transfers occurred in late 2005, triggered by the announcement that step-up would be removed for stock on January 1, 2006. Transfers made prior to January 2006 would still be subject to stepped-up basis treatment. The dotted red line before 2005 indicates the announcement of step-up's impending removal for stock. The solid red line before 2006 indicates the implementation of step-up's removal for stock.

check that a design not built on age recovers a similar aggregate. We do not use the under-50 UCG estimates: their event study shows no post-reform break, and the positive binary estimate reflects selection rather than a response to the reform.

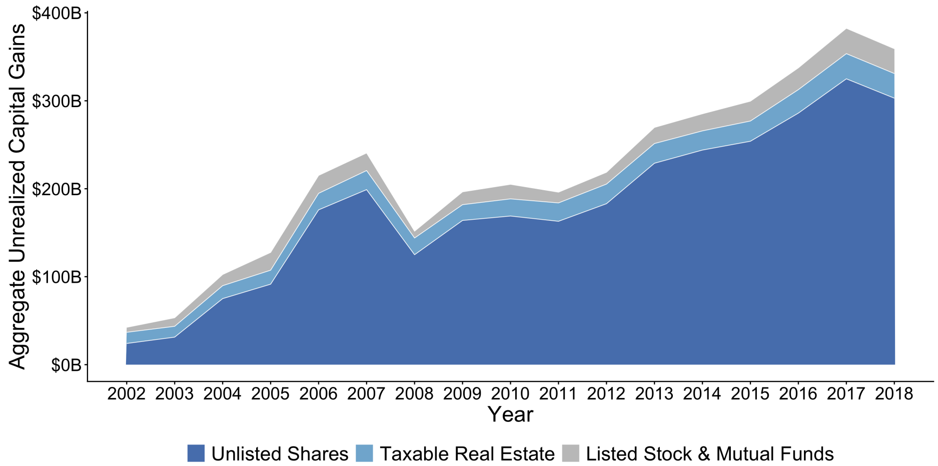
5.3 Interpreting the Dynamics: Anticipation and the Long Run

The two designs of Sections 5.1 and 5.2 produce a common, and at first surprising, finding: the medium-run coefficients are not smaller than the short-run responses. This lack of a short-term spike can be explained by the particular circumstances of Norway's removal of step-up once anticipation is taken seriously. Under an unanticipated removal of step-up (no early transfers), investors will realize many capital gains in the short-run in order to rebalance to the lower equity holdings appropriate to a carryover regime. But Norway's removal was anticipated: Norwegians could use the year-long lag between announcement and implementation to make *inter-vivos* transfers that still benefited from step-up. Capital gains eliminated via anticipatory transfers are not realized and therefore reduce the expected short-run realizations spike.[46]

If anticipatory transfers in 2005 absorbed gains that would otherwise have been realized (and would have otherwise caused a short-term spike), we should see an outlier volume of intergenerational stock transfers in the months between the December 2004 announcement and the January

[46]We omit (for length) a model extension that allows for anticipatory transfers. Simulations of this extended model confirm the intuition that the size of the short-term realization spike varies with the amount of anticipatory transfers and are available upon request.

Figure 6: Unrealized Capital Gains Over Time, 2002-2018



Note: This figure shows aggregate unrealized capital gains in taxable asset categories at the end of each year from 2002 (the initialization year enabled by the last pre-2003 transaction records) until 2018. Aggregate unrealized capital gains have mostly grown over time, with the exception of a sharp drop during the Great Recession. Trends are driven by unrealized capital gains in unlisted stock. The "Listed Stock & Mutual Funds" category in this graph includes the upper bound assumption about unrealized capital gains in mutual funds.

2006 implementation. Figure 5 documents exactly this: inheritance transfers of stock spiked in the second half of 2005, far above any level observed before or after. Contemporary press accounts urged Norwegians — particularly private business owners — to consider early inheritance transfers (Amundsen, 2005; Vigdal, 2005), and the data show the advice was widely taken up.

6 Aggregate Implications

Sections 6.1 and 6.2 clarify that the removal of step-up has two contrasting aggregate consequences: a large response in capital gains realization flows but a small change to the stock of unrealized capital gains. Section 6.3 discusses the external validity of our findings.

6.1 Aggregate Unrealized Capital Gains in the Post-Period

Above, we documented an increase in realizations of highly-appreciated stock by the original owners. A natural question is whether this increase translates into a meaningful reduction in the aggregate stock of unrealized capital gains after 2005. Figure 6 shows that it does not: the stock of unrealized capital gains in stock and taxable real estate showed sustained growth in our post-period, despite the higher level of taxable realizations. This pattern, which might initially seem at odds with our realization results, is in fact consistent with the relative magnitudes of the stock and the flow. An accounting identity for the year-to-year change in unrealized capital gains is:

$$UCG_{t+1} = UCG_t + AccCG_t - RCG_t - (1 - \theta_t)ICG_t \tag{6}$$

where $AccCG_t$ is net capital gains accrued during year t from appreciation of existing holdings and acquisition of new positions, RCG_t is taxable realizations, θ_t is an indicator for being in a carryover basis regime ($(1 - \theta_t)$ indicates a step-up regime), and ICG_t is capital gains transferred in inheritance.

Equation 6 makes clear that the year-to-year change in unrealized capital gains is the difference between a single inflow (net accrual) and two outflows (realization and inheritance). In 2004, both outflows were small relative to the stock: \$1.14 billion of taxable realizations and \$650 million of inherited capital gains, against a stock of \$91.92 billion in stock and taxable real estate (Table 1). Our estimated realization response (\$472 million) adds an additional outflow equivalent to 0.5% of the 2004 stock per year.[47]

To benchmark the inflow side of Equation 6, we apply a deliberately conservative real return of 5% per year to the \$146 billion of underlying asset value in 2004.[48] This generates an estimated $AccCG_t \approx \$7.3$ billion per year — more than four times the combined annual outflow from realization and inheritance. The implication is that, even after accounting for our estimated realization response, the stock of unrealized capital gains would be expected to grow by roughly \$5.05 billion (5.5%) per year through this period.[49]

A useful thought experiment is to consider what would have happened if the removal of step-up had *doubled* aggregate taxable realizations, rather than producing the actual increase we estimate. The additional \$1.14 billion of annual outflow would amount to 1.2% of the 2004 stock per year. Even under this extreme counterfactual, the implied net change in UCG_t would remain positive: at a 5% accrual rate, the stock of unrealized capital gains would still grow by roughly \$4.38 billion (4.8%) per year.

This stock-versus-flow comparison reconciles the two facts. Removing step-up generated a

[47]Note that our regression outcome includes RCG from all taxable asset classes, while RCG and ICG in Table 1 exclude mutual funds, whose gains we do not observe. Our comparisons here should therefore be taken as approximate and conservative.

[48]We view 5% as a conservative benchmark for two reasons. First, it is well below realized Norwegian listed-equity returns over the period we study: the Oslo Børs benchmark index more than doubled in real terms between 2004 and 2007. Second, holding $AccCG_t$ fixed at 5% applies the appreciation rate only to the 2004 asset value, ignoring growth in the asset base itself over time.

[49]Our calculated growth rate for UCG exceeds the assumed 5 percent accrual rate because appreciation accrues on all underlying asset value, while unrealized gains are only a portion of that total.

large response in capital gains tax revenue because taxable realizations are small relative to the underlying stock of accumulated, unrealized appreciation. The same arithmetic implies that even a much larger behavioral response would only modestly change the level of the UCG stock — the policy reform pulls forward gains that would otherwise have remained in the stock, but at a pace that is small relative to ongoing accrual. The realization response captures revenue from gains that would otherwise have been inherited tax-free, but it does not meaningfully shrink the stock of accumulated unrealized appreciation in the economy. For policymakers whose concern is the size of that stock — rather than the revenue raised on its eventual outflow — removing step-up is at best a partial instrument: the bulk of the stock is sustained by rising asset prices and the realization principle itself, not by step-up.

6.2 Aggregate Effects on Tax Revenue

6.2.1 Direct Effects on Capital Gains Tax Revenue

We convert our individual-level results into effects on aggregate capital gains realizations by calculating average treatment effects on the treated (see Tables 2 and 4) and multiplying that average by the total number of treated individuals. Our baseline results (Table 2) imply a short-run (2005-2008) average annual increase in realized capital gains of $472 million ($314 ATT $\times$ 1,503,089 Treated Individuals) against a counterfactual annual base of $1.80 billion (averaged 2005–2008). For the subset of individuals aged 50+, the UCG design estimates in Table 4 imply an annual rise in aggregate RCG of $500 million against a counterfactual annual base of $1.77 billion. This implies that the removal of step-up can be said to have increased capital gains tax revenue 26-28% above counterfactual.

6.2.2 Fiscal Externalities

We estimate that the behavioral response to step-up amplifies the mechanical loss of capital gains tax revenue: individuals defer more capital gains than they would in the absence of step-up, with no sign of reversal in our event studies. This amplification implies that step-up has a negative fiscal externality. Although two offsets in other tax bases work against this loss, the offsets cannot reverse step-up's negative fiscal externality unless the inheritance tax rate exceeds 50%.[50]

[50] The partially-offsetting tax effects are that, first, under carryover, the deferred gains may eventually be taxed when realized by the heirs. Second, an increase in capital gains deferrals increases the inheritance tax base. Appendix C quantifies the net fiscal externality using our model notation in a marginal value of public funds framework (Wildasin, 1984; Mayshar, 1990; Hendren, 2016). Our baseline MVPF is 0.84 and is below one in every variant we consider.

A negative fiscal externality distinguishes step-up from a capital gains rate cut, which would also benefit owners of appreciated assets but would reduce the return to deferral. Tax rate cuts typically have positive fiscal externalities (Hendren and Sprung-Keyser, 2020). This comparison points to a budget-neutral reform. Repealing step-up and returning the revenue as a reduction in the capital gains tax rate would benefit a similar (somewhat broader and somewhat less wealthy) population of capital gains owners, while reducing rather than amplifying distortions.[51] Given the concentration of step-up benefits in the top of the wealth distribution (Section 2) and the existence of the rate cut alternative, step-up is difficult to justify on either equity or efficiency grounds.

6.3 External Validity

Although our data and policy changes take place in Norway, we believe our findings are relevant to ongoing policy debates about the taxation of inherited capital gains in other countries. We find a large increase in taxable realizations when step-up is removed, which persists throughout our post-period, even after the inheritance tax is fully repealed in Norway. Because Norway granted step-up to *inter-vivos* transfers, the removal of step-up in Norway featured a year-long period during which owners of capital gains could make early inheritance transfers that would still benefit from step-up. The availability of this anticipatory response means that we find smaller short-run realization responses in Norway relative to what we would expect for the removal of step-up in other countries (such as the US or UK) which grant step-up to bequests at death but not to *inter-vivos* gifts.

There is also arguably less incentive to defer capital gains in Norway, relative to countries such as the US and UK. This smaller incentive is due to the fact that inter-corporate dividends and capital gains are largely tax exempt, which allows Norwegians to build up holding companies in which they can reallocate their portfolios without triggering the capital gains tax.[52] Step-up was correspondingly less valuable in Norway than in jurisdictions that tax intercorporate transactions. Furthermore, Norway levied an inheritance tax with low exemption thresholds during the step-up regime, meaning most people needed to pay inheritance tax to benefit from step-up. In contrast, the estate tax regimes currently in place in the US and UK exempt many inheritances from transfer tax while still granting them stepped-up basis. On the other hand, Norway's yearly wealth tax has been shown to encourage higher levels of savings (Ring, 2025).

[51]The revenue raised by repeal could finance a capital gains rate cut whose size depends on the elasticity of realizations within a carryover regime, which we expect to be smaller than the elasticity under step-up.

[52]Under Norway's exemption method (*fritaksmetoden*), corporate share gains are tax-exempt and qualifying dividends are largely exempt (only 3% is taxable).

As a final consideration, realization responses will differ across settings with different stocks of unrealized capital gains. Appendix Figure A5 compares our measure of pre-reform unrealized capital gains in Norway to the corresponding measure from the 2004 U.S. Survey of Consumer Finances, plotting average unrealized gains against average net wealth for groups defined by net worth percentile within each country. The two profiles lie close to one another throughout the Norwegian distribution: Norwegian households hold comparable unrealized gains to U.S. households with the same level of wealth. This comparison suggests that Norway is not an outlier in unrealized capital gains that could be drawn into realization by the removal of step-up. Because the gains exempted by step-up come overwhelmingly from the top of the wealth distribution (Section 2), the relevant region of Appendix Figure A5 is the upper tail, and there the U.S. wealth distribution extends past the range of the Norwegian distribution. A U.S. reform would thus apply to a base of unrealized gains at least as large as the one we study.

7 Conclusion

This paper provides the first quasi-experimental estimates of how the tax treatment of inherited capital gains affects realization behavior, using novel data on unrealized capital gains in Norway and the 2005-2006 removal of step-up. We document that unrealized capital gains are a large share of household wealth, especially at the top, and that the capital gains exempted under step-up represent a sizable fraction of the realization-based capital gains tax base. Two complementary research designs deliver estimates of a 26–28% increase in aggregate taxable realizations following the removal of step-up. The response is concentrated among the most-exposed ages and portfolios and persists through 2018. Removing step-up captures revenue from gains that would otherwise have been inherited tax-free, but it does not meaningfully shrink the underlying stock of unrealized capital gains, which is sustained by increasing asset prices and the realization principle itself. That the stock persists is not, however, an argument for retaining step-up: because step-up raises the return to deferral, it amplifies rather than offsets the lock-in distortion created by the capital gains tax, and a budget-neutral repeal that returned the revenue as a lower capital gains tax rate would benefit a similar population of capital gains owners while reducing distortions rather than increasing them.

References

Aaberge, Rolf, Marco Francesconi, Jørgen Modalsli, and Ola L. Vestad, "How Business Income Measures Affect Income Inequality and the Tax Burden," *Working Paper*, 2024.

Agersnap, Ole and Owen Zidar, "The Tax Elasticity of Capital Gains and Revenue-Maximizing Rates," *American Economic Review: Insights*, 2021, *3* (4), 399–416.

Alstadsæter, Annette and Erik Fjærli, "Neutral Taxation of Shareholder Income? Corporate Responses to an Announced Dividend Tax," *International Tax and Public Finance*, 2009, *16*, 571–604.

_ , Martin Jacob, Wojciech Kopczuk, and Kjetil Telle, "Accounting for Business Income in Measuring Top Income Shares: Integrated Accrual Approach Using Individual and Firm Data from Norway," *Journal of the European Economic Association*, 2026, *24.*

_ , Niels Johannesen, and Gabriel Zucman, "Tax Evasion and Inequality," *American Economic Review*, 2019, *109* (6), 2073–2103.

Amundsen, Toril Flate, "Ett år Igjen [One Year Left]," *Finansavisen*, 2005.

Auerbach, Alan J., "Capital Gains Taxation and Tax Reform," *National Tax Journal*, 1989, *42* (3), 391–401.

Auten, Gerald and David Joulfaian, "Bequest Taxes and Capital Gains Realizations," *Journal of Public Economics*, 2001, *81* (2), 213–229.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna, "Difference-in-Differences with a Continuous Treatment," *American Economic Review*, forthcoming.

Chen, Jiafeng and Jonathan Roth, "Logs with Zeros? Some Problems and Solutions," *The Quarterly Journal of Economics*, 2024, *139*, 891–936.

Ciani, Emanuele and Paul Fisher, "Dif-in-Dif Estimators of Multiplicative Treatment Effects," *Journal of Econometric Methods*, 2019, *8* (1), 1–10.

Dammon, Robert M., Chester S. Spatt, and Harold H. Zhang, "Optimal Consumption and Investment with Capital Gains Taxes," *The Review of Financial Studies*, 2001, *14* (3), 583–616.

DeMiguel, Victor and Raman Uppal, "Portfolio Investment with the Exact Tax Basis via Nonlinear Programming," *Management Science*, 2005, *51* (2), 277–290.

Dowd, Tim and Robert McClelland, "The Sensitivity of the Tax Elasticity of Capital Gains to Lagged Tax Rates and Migration," 2024.

Dybvig, Philip H. and Hyeng Keun Koo, "Investment with Taxes," 1996. Working paper, Washington University in St. Louis.

Eika, Lasse, Magne Mogstad, and Ola L. Vestad, "What Can We Learn about Household Consumption Expenditure from Data on Income and Assets?," *Journal of Public Economics*, 2020, *189*, 104163.

Fagereng, Andreas, Matthieu Gomez, Emilien Gouin-Bonenfant, Martin Holm, Benjamin Moll, and Gisle Natvik, "Asset-Price Redistribution," *Journal of Political Economy*, 2025, *133* (11).

Fairfield, Tasha and Michel Jorratt De Luis, "Top Income Shares, Business Profits, and Effective Tax Rates in Contemporary Chile," *Review of Income and Wealth*, 2016, *62* (S1), S120–S144.

Feldstein, Martin and Shlomo Yitzhaki, "The Effects of the Capital Gains Tax on the Selling and Switching of Common Stock," *Journal of Public Economics*, 1978, *9* (1), 17–36.

Gomes, Francisco, "Portfolio Choice over the Life Cycle: A Survey," *Annual Review of Financial Economics*, 2020, *12* (1), 277–304.

Gordon, Robert, David Joulfaian, and James Poterba, "Choosing between an Estate Tax and a Basis Carryover Regime: Evidence from 2010," *National Tax Journal*, 2016, *69* (4), 981–1002.

_ , _ , and _ , "Revenue and Incentive Effects of Basis Step-Up at Death: Lessons from the 2010 "Voluntary" Estate Tax Regime," *American Economic Review*, 2016, *106* (5), 662–667.

Gravelle, Jane G., "Capital Gains Tax Options: Behavioral Responses and Revenue," *Congressional Research Service Report*, 2026.

Hendren, Nathaniel, "The Policy Elasticity," *Tax Policy and the Economy*, 2016, *30* (1), 51–89.

_ and Ben Sprung-Keyser, "A Unified Welfare Analysis of Government Policies," *The Quarterly Journal of Economics*, 2020, *135* (3), 1209–1318.

Holt, Charles C. and John P. Shelton, "The Lock-In Effect of the Capital Gains Tax," *National Tax Journal*, 1962, *15* (4), 337–352.

Kiefer, Donald W., "Lock-In Effect within a Simple Model of Corporate Stock Trading," *National Tax Journal*, 1990, *43* (1), 75–94.

Larrimore, Jeff, Richard V. Burkhauser, Gerald Auten, and Philip Armour, "Recent Trends in US Income Distributions in Tax Record Data Using More Comprehensive Measures of Income Including Real Accrued Capital Gains," *Journal of Political Economy*, 2021, *129* (5), 1319–1360.

Lavecchia, Adam M. and Alisa Tazhitdinova, "Permanent and Transitory Responses to Capital Gains Taxes: Evidence from a Lifetime Exemption in Canada," *The Review of Economics and Statistics*, 2024, pp. 1–17.

Lorenz, M. O., "Methods of Measuring the Concentration of Wealth," *Publications of the American Statistical Association*, 1905, *9* (70), 209–219.

Mayshar, Joram, "On Measures of Excess Burden and Their Applications," *Journal of Public Economics*, 1990, *43* (3), 263–289.

Norwegian Ministry of Finance, "Skattereformen 2006 [Tax Reform 2006]," 2004. Accessed: 2024-10-14.

OECD, "Inheritance Taxation in OECD Countries," *OECD Tax Policy Studies*, 2021.

Poterba, James M., "Estate and Gift Taxes and Incentives for Inter Vivos Giving in the US," *Journal of Public Economics*, 2001, *79* (1), 237–264.

Ring, Marius A. K., "Wealth Taxation and Household Saving: Evidence from Assessment Discontinuities in Norway," *The Review of Economic Studies*, 2025, *92* (5), 3375–3402.

_ **, David G. Seim, and Gabriel Zucman**, "Personal Holding Companies, Tax Progressivity, and Inequality," *NBER Working Paper 35534*, 2026.

Smith, Matthew, Owen Zidar, and Eric Zwick, "Top Wealth in America: New Estimates under Heterogeneous Returns," *The Quarterly Journal of Economics*, 2023, *138* (1), 515–573.

Sørensen, Peter Birch, "Neutral Taxation of Shareholder Income," *International Tax and Public Finance*, 2005, *12* (6), 777–801.

Statistics Norway, "StatBank Table 07128: Dividends, capital gains and risk-free-return for individuals residing in Norway (closed series)," 2007. Accessed 20 July 2025.

The Norwegian Tax Administration, "Aksjonærmodellen [The Shareholder Model]," n.d. Accessed: 6 June 2025.

_ , "Examples showing how to complete the Shareholder register statement," [Accessed: 5 October 2024].

Thoresen, Thor O., Erlend E. Bø, Erik Fjærli, and Elin Halvorsen, "A Suggestion for Evaluating the Redistributional Effects of Tax Changes: With an Application to the 2006 Norwegian Tax Reform," *Public Finance Review*, 2012, *40* (3), 303–338.

Vigdal, Anne Marit, "Generasjonsskifte i Næring [Generational Succession in Business]," *Levanger-Avisa*, 2005. 19 February 2005.

Wildasin, David E., "On Public Good Provision with Distortionary Taxation," *Economic Inquiry*, 1984, *22*, 227–243.

Wolfson, Michael, Mike Veall, Neil Brooks, and Brian B. Murphy, "Piercing the Veil: Private Corporations and the Income of the Affluent," *Canadian Tax Journal/Revue fiscale canadienne*, 2016, *64* (1), 1–30.

Wooldridge, Jeffrey M., "Simple Approaches to Nonlinear Difference-in-Differences with Panel Data," *The Econometrics Journal*, 2023, *26* (3), C31–C66.

Yagan, Danny, "What is the Average Federal Individual Income Tax Rate on the Wealthiest Americans?," *Oxford Review of Economic Policy*, 2023, *39* (3), 438–450.

Appendix

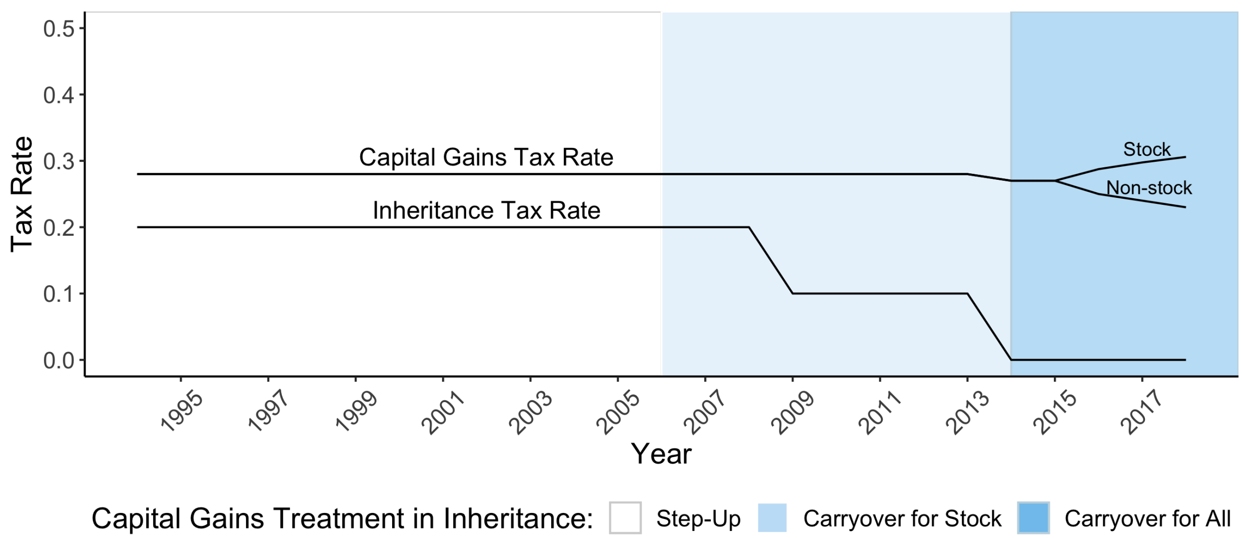
A Additional Appendix Tables & Figures

Table A1: Data Sources and Their Role in the Measurement Exercise

Data source	Coverage	Unit of observation	Key variables	Role in measurement
Panel A. Core administrative records				
Income tax records	1992–2018	Individual-year	Taxable income, Realized capital gain	Outcome variable and validates the transaction-based stock ledger.
Wealth tax records	1992–2018	Household-year	Assets + liabilities by asset category	Measures household net wealth, portfolio shares, and mutual-fund wealth.
Stock transactions	2003–2018 + last pre-2003 transaction	Individual-security-transaction	Asset ID, buyer/seller, quantity, price, transfer type	Reconstructs stock holdings, basis, RCG, UCG, and ICG.
Real estate transactions	1993–2018 + last pre-1993 transaction	Individual-property-transaction	Property ID, buyer/seller, price, transfer type	Reconstructs real-estate holdings, basis, RCG, UCG, and ICG.
Panel B. Valuation and validation inputs				
Unlisted firm accounts (financial and tax-accounting records)	1993–2018	Firm-year	Sales, income, book values, industry, ownership links	Constructs market value estimates for unlisted firms.
Shareholder Registry	2004–2018	Individual-company-year	End-of-year shareholdings	Validates the transaction-based stock ledger.

Note: This table summarizes the main data sources used in the measurement exercise. Listed stock values are obtained from Thomson-Reuters Datastream. Real estate values use the property-level market-value estimates from Eika et al. (2020). Mutual fund wealth is observed in the wealth data, but mutual fund transactions and basis are not observed; we therefore bound mutual-fund UCG as described in Section 1.3. Stock and real estate transaction records include sales, inheritance transfers, and other transfer types. Data limitations are discussed in Section 1.3.

Figure A1: Capital Gains and Inheritance Policy in Norway, 1994–2018



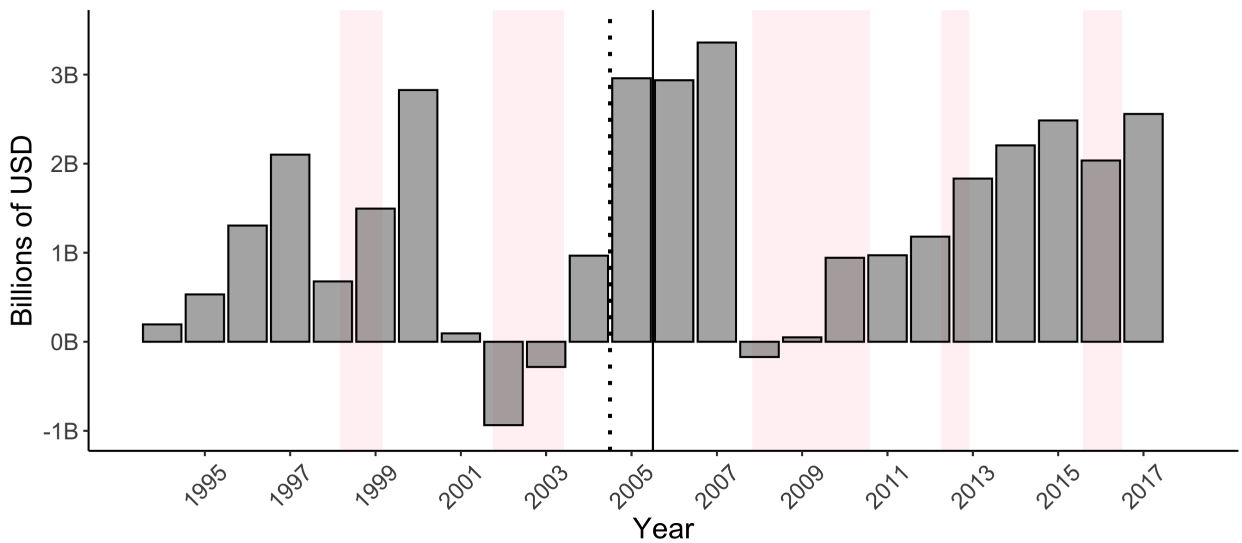
Note: This figure summarizes the changes to capital gains and inheritance taxation that took place during our study period. The statutory capital gains tax rate on all taxable assets was steady at 28% until 2014. From 2016, separate capital gains tax rates were introduced for stock and non-stock assets. Throughout our period, capital gains taxation in Norway has had a flat rate structure, although the rate was made to depend on the asset type (stock or non-stock) in later years. In contrast, inheritance tax rates (prior to their abolition) were progressive in the cumulative amount of the inheritance received, and depended on the relationship between giver and recipient. This graph depicts the top statutory inheritance tax rate for closely related family members, which fell from 20% to 10% in 2009 and to 0% with the abolition of the inheritance tax in 2014. See Appendix Table A2 for more information on inheritance tax brackets. The shading of the graph depicts changes to the treatment of inherited capital gains. Prior to 2006, all gifted and inherited assets benefited from stepped-up basis. From 2006, gifted and inherited stock with positive capital gains were switched to a system of carryover basis. From 2014, all gifted and inherited assets carried their basis over to the recipient, unless the asset would not have been subject to capital gains taxation in the hands of the original owner.

Table A2: Inheritance Tax Schedule, 1985-Present

Years	Amount Inherited (NOK)	Tax Rate for:		
		Spouses	Close Relatives	Other Recipients
1985-1998	0 - 100,000	0	0	0
	100,000 - 400,000	0	0.08	0.10
	400,000 +	0	0.20	0.30
1999-2002	0 - 200,000	0	0	0
	200,000 - 500,000	0	0.08	0.10
	500,000 +	0	0.20	0.30
2003-2008	0 - 250,000	0	0	0
	250,000 - 550,000	0	0.08	0.10
	550,000 +	0	0.20	0.30
2009-2013	0 - 470,000	0	0	0
	470,000 - 800,000	0	0.06	0.08
	800,000 +	0	0.10	0.15
2014-present	Any	0	0	0

Note: This table depicts the Norwegian inheritance tax scheme from 1985 until the inheritance tax was abolished in 2014. Inheritance tax was levied without distinction between *inter vivos* gifts and bequests at the time of death. The rate structure depended on the cumulative amount of the gift/inheritance received from the individual giver, and the relationship between the giver and the recipient. The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year. In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.

Figure A2: Aggregate Net Capital Gains Realizations by Year, 1994-2018



Note: This figure depicts the total amount of net capital gains (i.e., capital gains minus capital loss) realized in Norway from 1994 to 2018, in terms of 2018 USD. The light pink shading denotes recessionary periods, as classified by the OECD. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock.

Table A3: Realization Responses by Age Group & Specified Type of Capital Gain

Panel (a): Real Estate Capital Gains (2000-2004,2006-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	-0.088	-0.127	-0.064	-0.139	-0.098
	(0.100)	(0.095)	(0.115)	(0.103)	(0.082)
Number of Observations	22,003,105	22,003,105	22,003,105	22,003,105	22,237,262
Implied ATT	\$-23	\$-42	\$-16	\$-22	\$-26

Panel (b): Private Business Capital Gains (2003-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	-0.041	0.043	-0.156	0.594***	-0.013
	(0.161)	(0.178)	(0.332)	(0.162)	(0.127)
Number of Observations	17,034,859	17,034,859	17,034,859	17,034,859	17,268,802
Implied ATT	\$-23	\$21	\$-38	\$58	\$-5

Panel (c): Listed Stock, Mutual Funds & Other Capital Gains (2003-2004,2006-2008)

	(1)				(2)
Age Range of Treatment Group:	50-59	60-69	70-79	80-89	All 50+
Group × POST	0.224*	0.382***	0.603***	0.757***	0.352***
	(0.095)	(0.102)	(0.146)	(0.101)	(0.093)
Number of Observations	13,978,308	13,978,308	13,978,308	13,978,308	14,133,446
Implied ATT	\$176	\$272	\$289	\$219	\$225

Note: This table presents difference-in-difference estimates from our age-based empirical strategy (as in Tables 2 and 3), but with the outcome variable in each panel corresponding to particular types of capital gain. The population is income tax filers aged 30 and above. All regressions include covariates for whether the individual ever owns a private business and for their highest age-net worth bin. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the age group below the numerical column header, and equal 0 if the individual is in the 30-49 age group. Other age groups are not included. The indicator "POST" equals 1 for years after 2004, and 0 before then. Standard errors clustered by birth cohort are presented in parentheses below. ATT values are in 2018 USD. Capital gains for "Real Estate" are not observed in 2005 due to a data issue. Capital gains in "Private Business" and "Listed Stock, Mutual Funds + Other" are only observed from 2003 (when our stock transactions data begins). All analyses in this table end in 2008, before the 2009 decrease in inheritance tax rates. Individuals in these regressions are classified as private business owners if, since the age of 25, anyone in their household owned unlisted stock between 1993 and 2018. Individuals are assigned their household's highest net wealth percentile over the same period of time. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

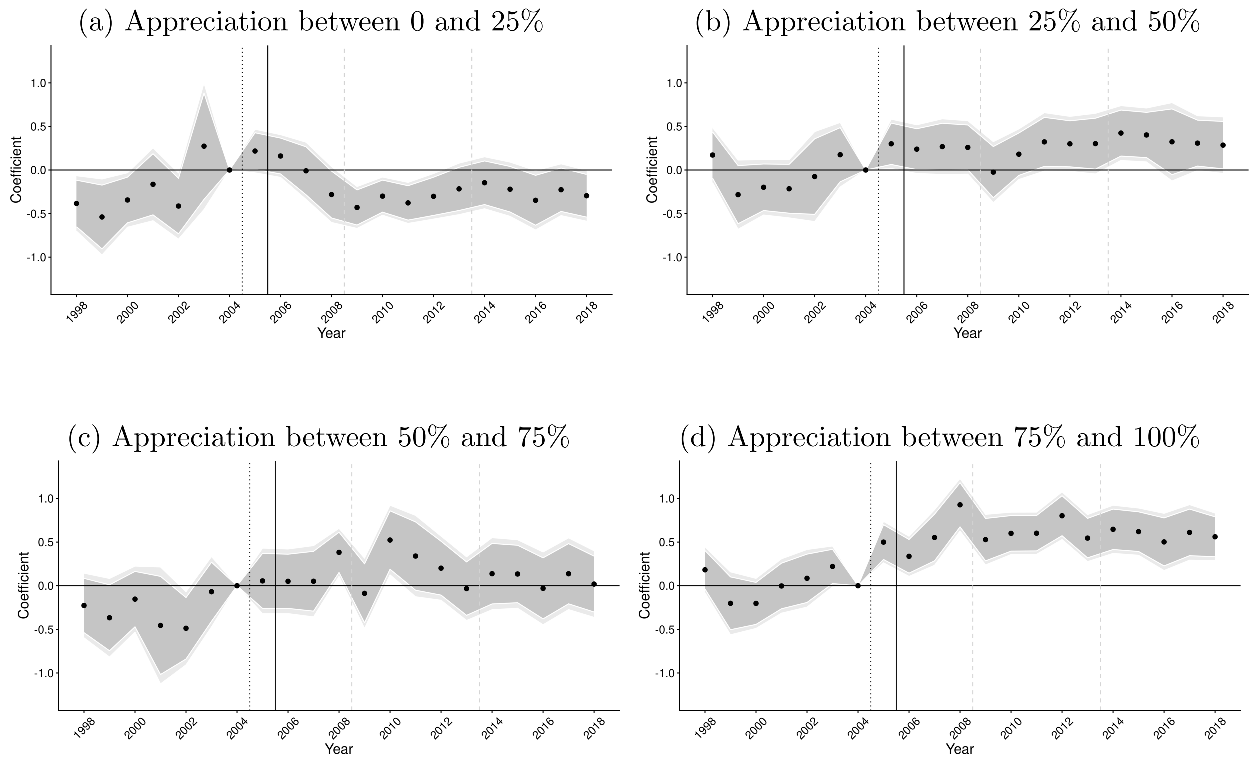
Table A4: Medium-Run Realization Responses from Age-based Empirical Design

Difference-in-Difference Regressions (2000-2004 vs 2009-2013)

	Specification:		*Implied Treatment Effects:*	
Ages in Treatment Group:	(1)	(2)	%Δ	ATT
All Aged 50+		0.309^{**}	36%	$213
		(0.113)		
Ages 50-59	0.100		11%	$81
	(0.132)			
Ages 60-69	0.453^{***}		57%	$359
	(0.118)			
Ages 70-79	0.588^{***}		80%	$307
	(0.131)			
Ages 80-89	0.546^{***}		73%	$174
	(0.144)			
Number of Treated Individuals	1,566,534	1,601,469		
Number of Total Observations	28,358,517	28,688,043		

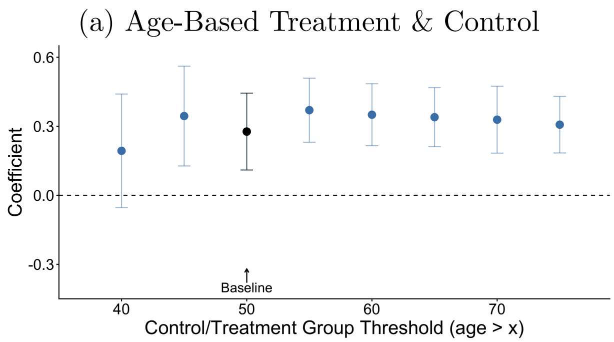
Note: This table presents an alternative version of Table 2 where the post-period is defined from 2009–2013 (rather than from 2005–2008). The outcome variable is net realized capital gains from any asset class, as reported yearly on each individual's tax return. An observation is an individual-year income tax filing in 2000–2004 or 2009–2013 and we include all income tax filers aged 30 and above. Mutually exclusive treatment indicator(s) equal 1 if the individual is in the indicated age group and equal 0 if the individual is in the 30-49 age group. Standard errors clustered by birth cohort are presented in parentheses (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). All regressions control for an indicator for private business ownership and a bin of highest-ever household net worth. Individuals in these regressions are classified as private business owners if, since the age of 25, anyone in their household owned unlisted stock between 1993 and 2018. Individuals are assigned their household's highest net wealth percentile over the same period of time. See Figure 4 for coefficients from this sample broken out by age decile rather than 10-year age bucket.

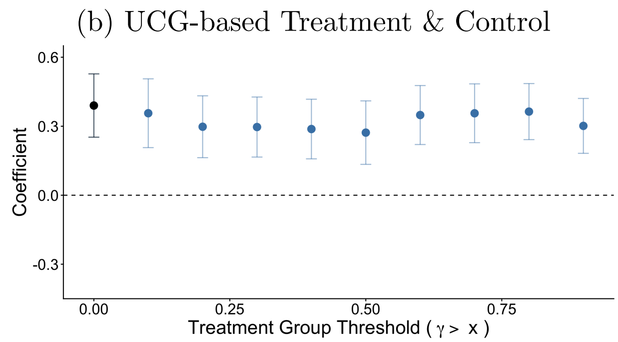
Figure A3: Event Studies for the UCG-based Design, by Bin of Treatment Intensity



Note: This figure shows heterogeneity in the average responses to the removal of stepped-up basis for inherited stock in our alternative UCG-based empirical design. Panel (a) depicts event study coefficients for individuals whose stock holdings, at the end of 2004, have a ratio of unrealized capital gains to value (γ_{2004}) between zero and 0.25. Panel (b) shows the corresponding figure for individuals with 2004 stock holdings with γ_{2004} between 0.25 and 0.5. Panel (c) is the corresponding figure for those with γ_{2004} between 0.5 and 0.75. Panel (d) is the corresponding figure for those with γ_{2004} between 0.75 and 1. In all panels, the control group is individuals whose stock portfolios in 2004 had zero capital gains or net capital losses and the outcome variable is net realized capital gains from any asset class. All regressions include controls for age group, 2004 equity share, private business ownership, and age-specific rank in the household wealth distribution. Coefficients in Panel (a) should be interpreted with caution: membership in this bin is the most sensitive to transitory return shocks, since it borders the control group's threshold of zero, and by construction these individuals hold only small unrealized gains relative to portfolio value. The negative coefficients after 2007 are consistent with mean reversion on both sides of this threshold: a small stock of unrealized gains is quickly exhausted by early realizations, while control-group individuals' positions appreciate over time, giving them gains to realize in later years. Panels (b)–(d), whose bins lie away from the boundary, show effects that increase with treatment intensity γ_{2004}. The light (dark) grey ribbons depict 95% (90%) confidence intervals from standard errors clustered by birth cohort. The dotted black line before 2005 indicates the announcement of step-up's impending removal for stock. The solid black line before 2006 indicates the implementation of step-up's removal for stock. The dashed gray line before 2009 indicates the cut to the statutory inheritance tax rate. The dashed gray line before 2014 indicates the full removal of the inheritance tax, and the beginning of a period of changing capital gains tax rates.

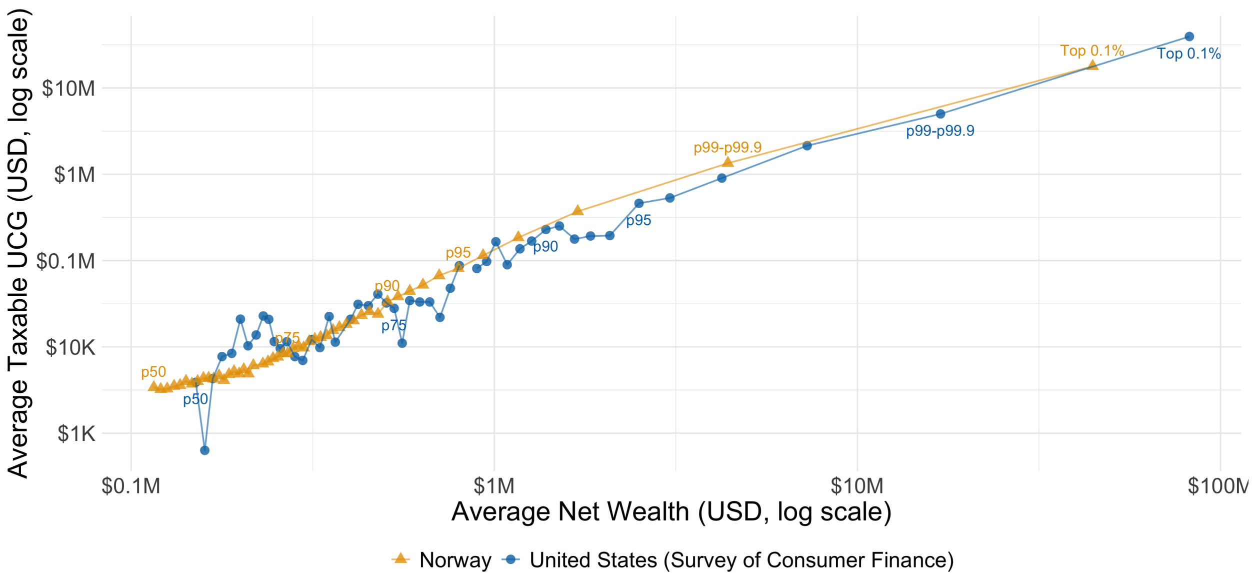
Figure A4: Robustness to Alternative Control Group Definitions





Note: This figure shows how our binary estimates of realization responses from Tables 2 and 4 change when we alter the threshold at which we define our "treatment" and "control" groups. Each point is a difference-in-difference estimate of the binarized specification of Equation 4 with the treatment variable defined as an indicator variable for age or γ_{2004} above the given threshold. The threshold in Panel (a) is defined in terms of age (baseline threshold: age 50+ years) and the threshold in Panel (b) is defined in terms of the individual's unrealized capital gain ratio (γ_{2004}) at the end of 2004 (baseline threshold: $\gamma_{2004} > 0$). As discussed in Section 5, both of our empirical designs are "continuous" or "dose-response" difference-in-differences designs, and neither has an obvious cutoff threshold at which to cleanly separate the treatment and control group. Even when true treatment effects increase monotonically with the running variable, incremental increases in the choice of threshold may increase or decrease estimated effects. Increasing the treatment threshold in the treated range misattributes some of the true causal effect to the control group mean but also re-defines the treatment group to only include more intensely treated members. Which mechanism dominates at intermediate values will depend on the slope of the heterogeneous treatment effects by dosage.

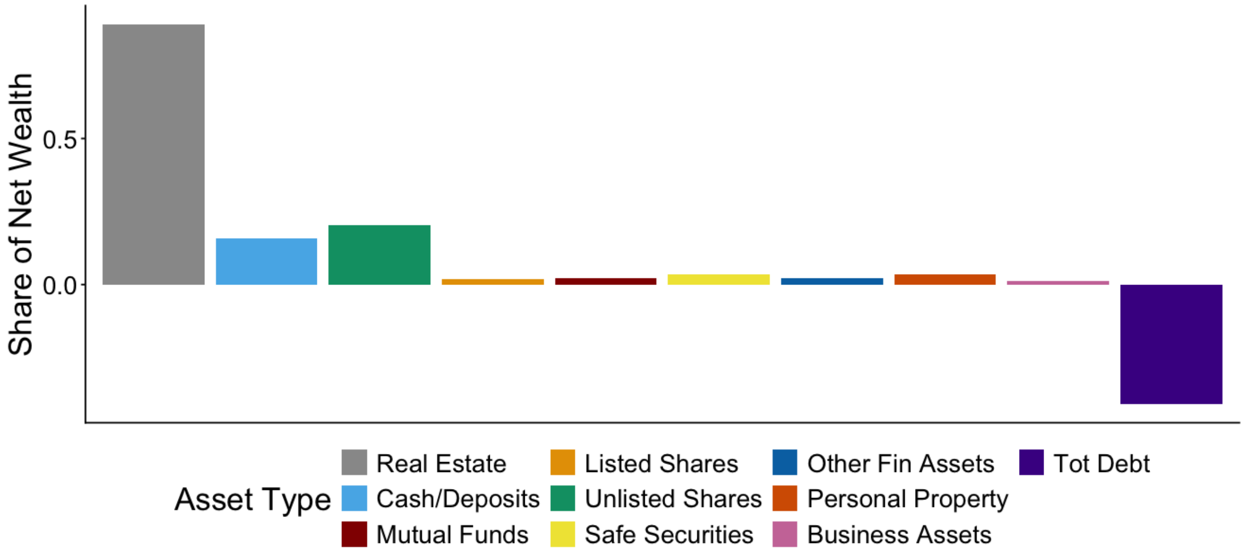
Figure A5: UCG vs. Net Wealth: Comparing Norway and the United States, 2004



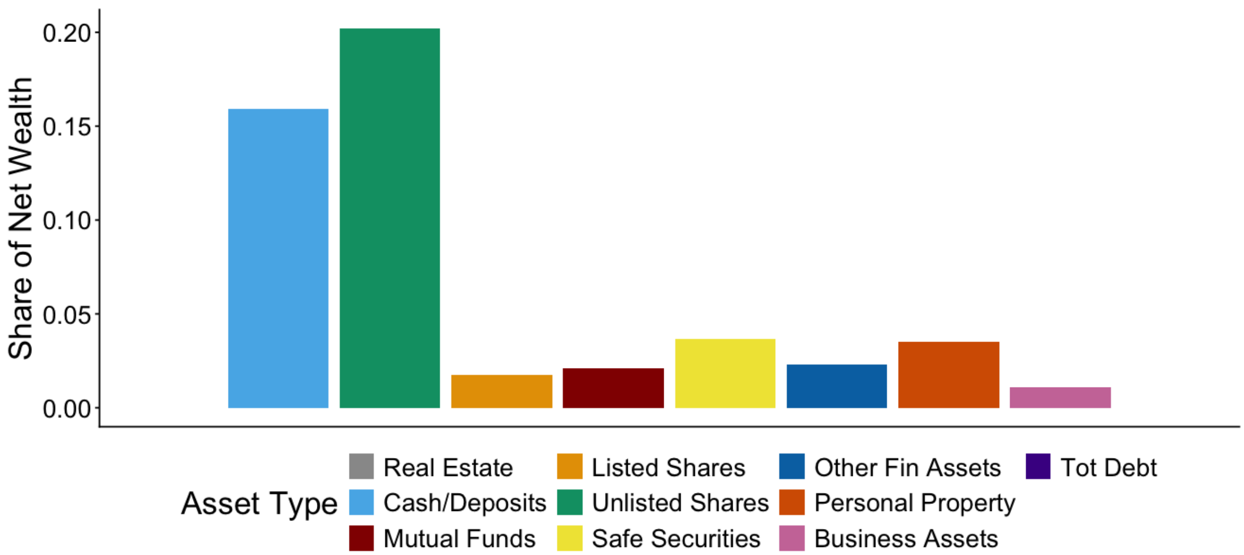
Note: This figure compares our measure of unrealized capital gains in taxable asset categories (i.e., excluding non-taxable real estate) in Norway at the end of 2004 to unrealized capital gains in categories other than primary residences reported in the 2004 U.S. Survey of Consumer Finances. Each point in the figure represents a mean for individuals grouped (within their respective data source) by total net worth. We group individuals at integer percentiles of net worth below the 99th percentile, but then split the top 1% into a group below the 99.9th percentile (labeled "p99-p99.9") and a group of the Top 0.1%. Upper-middle-class Norwegians have comparable but slightly higher levels of unrealized capital gains than middle-class U.S. residents with the same amount of total wealth. Average unrealized capital gains converge for U.S. and Norwegian millionaires. At the very top, the top of the U.S. wealth distribution extends past the range of the Norwegian distribution, so it is not possible to directly compare magnitudes.

Figure A6: Average Portfolio Shares in 2004

(a) All Asset Types



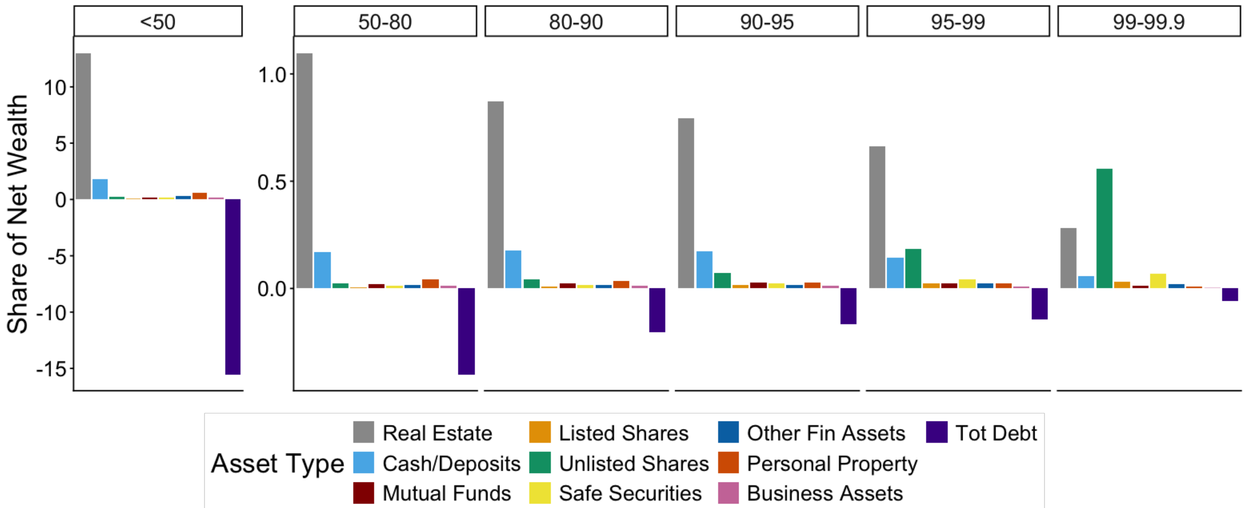
(b) Without Real Estate or Debt



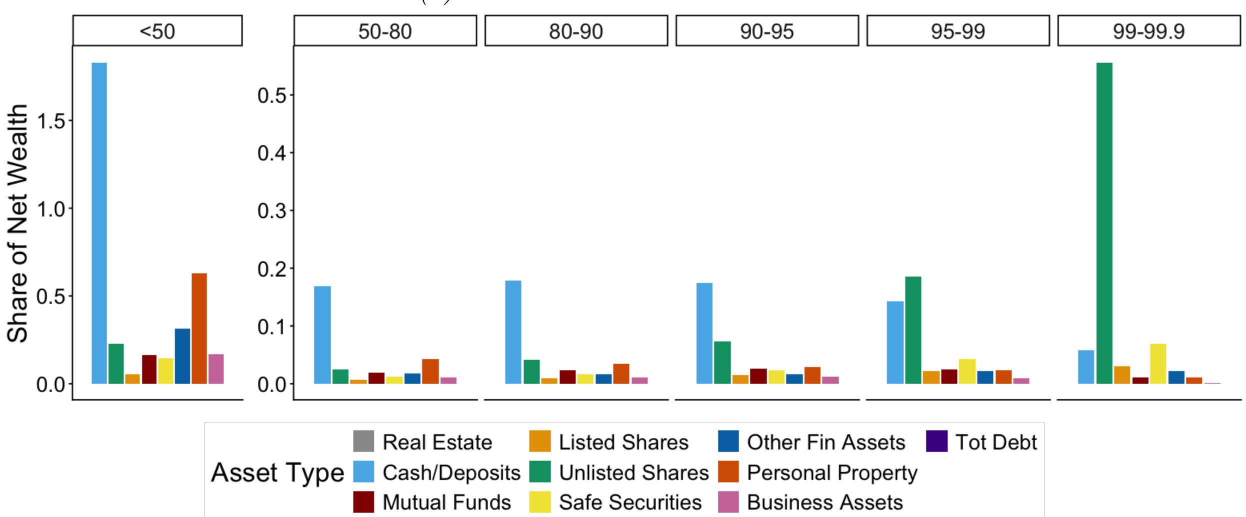
Note: This figure depicts portfolio shares for the average Norwegian household at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt.

Figure A7: Average Portfolio Shares in 2004, by Net Worth Percentile

(a) All Asset Types



(b) Without Real Estate or Debt



Note: This figure depicts portfolio shares for Norwegian households broken out by percentile of net worth at the end of 2004. Panel (a) presents all asset types. Panel (b) is a zoomed in version for those asset types other than real estate and debt. Note that averages below the 50th percentile include people with negative net worth (lots of debt), implying portfolio shares greater than 1. Unlisted stock is a disproportionately important part of very rich people's portfolios.

B Model Simulation

This section provides implementation details of how we numerically solve & simulate the model from Section 4.

B.1 Model Simulation Details

B.1.1 Parameterizations

We parameterize the utility function for real consumption $\tilde{c}_t = \frac{c_t}{(1+i)^t}$ and bequests $\tilde{B}_t = \frac{B_t}{(1+i)^t}$:

$$u(\tilde{c}_t) \equiv \tfrac{1}{1-\rho}\tilde{c}_t^{\,1-\rho} \qquad \nu(\tilde{B}_t) \equiv \tfrac{\beta}{1-\beta}\tfrac{1}{1-\rho}\big(A \cdot \tilde{B}_t\big)^{1-\rho}$$

where β is the discount factor, ρ is the relative risk aversion coefficient and $A \equiv \frac{r-i}{(1+i)}$ is the individual's utility weight on bequests for constant inflation rate i. Note that this formulation of the utility over bequests is equivalent to assuming that the proceeds of the inheritance are used to purchase a perpetual annuity which provides the heir with a stream of consumption $A \cdot \frac{B_{\mathcal{T}}}{(1+i)^{\mathcal{T}}}$, where A is the after-tax real perpetuity rate.

B.1.2 Recursive Formulation

We can write the Bellman for state variables $X_t = [P_t, U_t, n_{t-1}, W_t]'$ as

$$V_t(X_t;\theta) = \mathbb{P}(\mathcal{T} > t | \mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \big\{ u\big(\frac{c_t}{(1+i)^t}\big) + \beta E_t[V_{t+1}(X_{t+1};\theta)] \big\} + \mathbb{P}(\mathcal{T} = t | \mathcal{T} \geq t)\nu\big(\frac{B_t}{(1+i)^t}\big) \quad s.t.$$

$$W_{t+1} = \big(1+r\big)\big(W_t - n_t P_t - c_t - \tau_c R_t\big) + n_t P_{t+1}$$

$$B_t = W_t(1-\tau_I) - \theta\tau_c U_t^+$$

B.1.3 Normalized Formulation

Our choice of parameterization (with constant relative risk-aversion) implies that optimal consumption and asset positions scale with wealth. Following Dammon et al. (2001)'s methodology, we solve a transformation of the above model that normalizes all of the variables by beginning-of-period wealth. With beginning-of-period wealth W_t as the numeraire, we can write a normalized value function $v_t(x_t;\theta) \equiv V_t(X_t;\theta)/\big(W_t/(1+i)^t\big)^{1-\rho}$,

$$v_t(x_t;\theta) = \mathbb{P}(\mathcal{T} > t | \mathcal{T} \geq t) \max_{c_t, n_t \geq 0} \Big\{ u\big(\tfrac{c_t}{W_t}\big) + \beta E_t\big[v_{t+1}(x_{t+1};\theta)\big(\tfrac{W_{t+1}}{W_t(1+i)}\big)^{1-\rho}\big] \Big\} + \mathbb{P}(\mathcal{T} = t | \mathcal{T} \geq t)\, \tilde{\nu}(x_t;\theta)$$

with the normalized death payoff:

$$\tilde{\nu}(x_t;\theta) = \frac{\beta}{1-\beta}\frac{1}{1-\rho}\big(A(1-\tau_I-\theta\tau_c s_t\gamma_t^+)\big)^{1-\rho}$$

This formulation only requires two (normalized) state variables: $x_t = [s_t, \gamma_t]$, where $s_t \equiv \frac{n_{t-1}P_t}{W_t}$ is the beginning-of-period equity share of wealth and $\gamma_t \equiv U_t/s_tW_t$ is the beginning-of-period unrealized capital gains share of equity. $\gamma_t^+ \equiv \max\{0, \gamma_t\}$.

B.1.4 Simulation

The model cannot be solved analytically, so we use numerical techniques to solve for the policy and value functions. We employ our (numerical) policy and value functions in a simulation which assumes that every individual starts life at age 20 ($t = 0$) with $W_0 = 1$ and no unrealized capital gains ($U_0 = 0$). Individuals live up to age 100 ($T = 80$) and face a yearly probability of death based on official Norwegian mortality numbers (Statistics Norway, 1966–2024). Table A5 displays our chosen values that further parameterize the model for simulation purposes. At advanced ages the value function is nearly flat in the portfolio choice (within 3×10^{-4} of the optimum over a ± 0.05 range of s), so the solved policy is identified only up to numerical noise within this indifference set; for figures we select from that set via a centered three-age moving average, which changes lifetime value by less than the indifference tolerance and mean realizations by under 1%. Note that borrowing in the risk-free asset is permissible in our model. However, since we adopt the standard convention that our CRRA utility functions u and ν take the value $-\infty$ outside the strictly positive domain, $W_{t+1} \leq 0$ is never chosen. Additionally, our numerical state grid for s_t restricts the equity position after trading to $n_tP_t \leq W_t$, which, by the budget constraint, further restricts maximum borrowing.

Table A5: Parameters for Model Simulation

Parameter	Symbol	Value	Source	Timeframe
Financial markets				
After-tax risk-free rate	$r \equiv (1-\tau_d) r_f$			
Pretax risk-free rate	r_f	0.0376	Fagereng et al. (2017) (using Klovland (2004))	1900-2005
Mean equity return	$\bar{g}$	0.0697	Fagereng et al. (2017) (using Dimson et al. (2008), Klovland (2004))	1900-2005
Std. dev. of equity return	σ	0.244		
Inflation rate	i	0.023	Grytten (2004)	1995-2003
Tax rates				
Tax rate on interest	τ_d	0.28	Norwegian Law (flat rate)	1992-2013
Tax rate on realized capital gains	τ_c	0.28	Norwegian Law (flat rate)	1992-2013
Inheritance tax rate	τ_I	0.20	Norwegian Law (top rate)	1985-2008
Preferences				
Discount factor	β	0.96		
Coefficient of relative risk aversion	ρ	3		
Lifecycle				
Starting age	$t=0$	20		
Maximum periods	T	80		
Wealth at starting age	W_0	1		
Unrealized Gains at start	γ_0	0		
Mortality data source			Statistics Norway (1966–2024)	2004

Note: This table displays our chosen values that parameterize the model for simulation purposes. The "Financial markets" parameters are chosen to follow Norwegian portfolios in the decades prior to our policy change and assume that Norwegian equity portfolios are weighted 80% to Norwegian equity, 20% to global equity (following Fagereng et al. (2017)). We do not model dividends, so the entire equity return accrues as capital gains. "Tax rates" are chosen to be consistent with actual Norwegian law around the time of our policy change. Table A6 provides additional detail on how we transform the equity-related parameters from the literature into the (nominal) values listed here. Table A2 provides additional details on inheritance tax rates.

Table A6: Calculation of Parameters for Model Simulation

Parameter	Real Value	Nominal Conversion	Source
Mean Equity Returns			
Norway	0.0428		Dimson et al. (2008)
World	0.0575		Dimson et al. (2008)
80/20 Weighted Portfolio	0.0457	0.0697	Fagereng et al. (2017)
Std. dev. of Equity Returns			
Norway	0.2696		Dimson et al. (2008)
World	0.1723		Dimson et al. (2008)
corr(Norway, World)	0.61		Ødegaard (2007)
80/20 Weighted Portfolio	0.238	0.244	Fagereng et al. (2017)

Note: This table explains how we transform values taken from the literature into the values used in our model simulation. We employ Fagereng et al. (2017)'s estimate (combining administrative data on directly-held stock with aggregate data on mutual fund holdings) that Norwegian equity portfolios including mutual funds are weighted 80% Norway, 20% the rest of the world. Real values are transformed into nominal values using an inflation rate of 2.3% which Grytten (2004) finds to be the average inflation rate in Norway for 1995-2003. Norges Bank adopted an inflation target of 2.5% in 2001.

C MVPF Derivation

C.1 MVPF Derivation for a Marginal Policy Change $d\theta$

We derive our equation for the MVPF under a marginal policy change (Equation 7) from the decision of an individual entering the period before death with positive unrealized capital gains ($U_{\mathcal{T}-1} > 0$). Assuming that this individual makes his final choice of stock holdings $n_{\mathcal{T}-1} \in [0, n_{\mathcal{T}-2}]$ and that his heir realizes inherited gains immediately upon receipt (before any additional price appreciation), we can simplify the terminal period budget constraint (from Equation 1) to $W_{\mathcal{T}} = W_{\mathcal{T}-1} - c_{\mathcal{T}-1} - \tau_c U_{\mathcal{T}-1}(1 - \frac{n_{\mathcal{T}-1}}{n_{\mathcal{T}-2}})$.

Then for any $\theta \in [0, 1]$ (i.e., step-up, carryover, or anything in between), we can write affected government revenue as:

$$\begin{aligned} G(\theta) &= \tau_I\big(W_{\mathcal{T}}(\theta) - 0.2\,\theta\,U_{\mathcal{T}}(\theta)\big) + \theta\tau_c U_{\mathcal{T}}(\theta) + \tau_c(1 - \tfrac{n_{\mathcal{T}-1}(\theta)}{n_{\mathcal{T}-2}})U_{\mathcal{T}-1} \\ &= \tau_I W_{\mathcal{T}}(\theta) + \theta(\tau_c - \kappa)U_{\mathcal{T}}(\theta) + \tau_c(1 - \tfrac{n_{\mathcal{T}-1}(\theta)}{n_{\mathcal{T}-2}})U_{\mathcal{T}-1}, \qquad \kappa \equiv 0.2\tau_I \end{aligned}$$

The $-0.2\,\theta\,U_{\mathcal{T}}$ term inside the inheritance-tax base implements Norway's valuation deduction (Section 3): heirs of carried-over gains had their inheritance-tax values reduced by 20% of the carried-over gain, so each dollar of gain transferred under carryover generates an inheritance-tax credit of

$\kappa = 0.2\tau_I$. Plugging in $U_{\mathcal{T}-1}\frac{n_{\mathcal{T}-1}(\theta)}{n_{\mathcal{T}-2}} = U_{\mathcal{T}}(\theta)$, we differentiate $G(\theta)$ with respect to θ to recover the change in net government revenue for any marginal policy change $d\theta$:

$$dG(\theta) = \tau_I dW + (\tau_c - \kappa)U_{\mathcal{T}}(\theta)d\theta + \big((\tau_c - \kappa)\theta - \tau_c\big)\frac{U_{\mathcal{T}-1}}{n_{\mathcal{T}-2}}dn$$

Define $R_{\mathcal{T}-1} \equiv U_{\mathcal{T}-1} - U_{\mathcal{T}}$. From Equation 3, this implies $\frac{dR_{\mathcal{T}-1}}{d\theta} = -\frac{U_{\mathcal{T}-1}}{n_{\mathcal{T}-2}}\frac{dn_{\mathcal{T}-1}}{d\theta}$. Using this to substitute in for dn and the per-period budget constraint to substitute in for dW, we get:

$$dG(\theta) = -\tau_I dc + (\tau_c - \kappa)U_{\mathcal{T}}(\theta)d\theta + \big(\tau_c(1 - \theta - \tau_I) + \kappa\theta\big)\frac{dR_{\mathcal{T}-1}}{d\theta}d\theta$$

This is the denominator of the MVPF for a marginal policy change. We will additionally assume $dc \approx 0$ (i.e., the older generation's marginal propensity to consume out of a terminal period wealth shock is zero).[53] The numerator of the MVPF is the dynasty's willingness-to-pay to avoid carryover policy, which is equal to their tax savings under step-up: the heir's capital gains tax avoided, net of the inheritance-tax deduction forgone, $WTP(\theta) = (\tau_c - \kappa)U_{\mathcal{T}}(\theta)d\theta$. Divide through to get $MVPF_{d\theta} = \frac{WTP(\theta)}{dG(\theta)} = \frac{1}{1+FE_{d\theta}}$:

$$MVPF_{d\theta} = \frac{1}{1 + FE_{d\theta}}; \qquad FE_{d\theta}(\theta) = \left(\frac{\tau_c(1 - \theta - \tau_I) + \kappa\theta}{\tau_c - \kappa}\right)\frac{dR_{T-1}/d\theta}{U^+_{\mathcal{T}}(\theta)} \tag{7}$$

Each marginal dollar of induced realization R raises capital gains revenue by τ_c today, forgoes the $\theta\tau_c$ that the heir would have paid on those gains, and forgoes $\tau_I\tau_c$ of inheritance tax because the capital gains tax paid on induced realizations shrinks the taxable estate. Norway's 20% deduction on the inheritance tax for carried-over capital gains prior to 2014 enters twice: it lowers the mechanical revenue per dollar of carried-over gain from τ_c to $\tau_c - \kappa$, and it returns $\kappa\theta$ per dollar of induced realization (since gains realized before death don't generate the credit). The denominator $U^+_{\mathcal{T}}(\theta)$, the annual flow of positive unrealized gains transferred at death, is the mechanical base of the tax expenditure.[54]

[53]If the older generation's MPC out of terminal-period wealth is positive, our assumption $dc \approx 0$ overstates the MVPF: a positive MPC of m would add $m\tau_I \cdot dR/d\theta$ to the denominator of the MVPF. Since our central conclusion is that the MVPF is small, $dc \approx 0$ is the conservative choice.

[54]Because Norway's carryover applied only to gains (losses were still stepped-up), our formula relies on positive inherited gains only.

C.1.1 Aggregation to the Discrete Reform

The removal of step-up is a discrete move from $\theta = 0$ to $\theta = 1$, so we aggregate the marginal effects by integrating along the policy path (Kleven, 2021). The MVPF of the reform is the ratio of the integrated willingness-to-pay to the integrated revenue change:

$$MVPF_{0\to1} \;=\; \frac{\displaystyle\int_0^1 (\tau_c-\kappa)U_{\mathcal{T}}(\theta)\,d\theta}{\displaystyle\int_0^1 \Big[(\tau_c-\kappa)U_{\mathcal{T}}(\theta) + \big(\tau_c(1-\theta-\tau_I)+\kappa\theta\big)\tfrac{dR_{\mathcal{T}-1}}{d\theta}\Big]\,d\theta}.$$

Following Kleven (2021), we approximate the $MVPF$ for the full removal of step-up by assuming linearity between endpoints for both willingness-to-pay and mechanical revenue, so that Equation 8 follows from the trapezoid approximations:

$$\int_0^1 (\tau_c-\kappa)U_{\mathcal{T}}(\theta)\,d\theta \approx (\tau_c-\kappa)\bar{U}, \qquad \int_0^1 \big(\tau_c(1-\theta-\tau_I)+\kappa\theta\big)\Delta R_{\mathcal{T}-1}\,d\theta \approx \Big(\tau_c\Big(\tfrac{1}{2}-\tau_I\Big)+\tfrac{\kappa}{2}\Big)\Delta R_{\mathcal{T}-1},$$

with $\bar{U} \equiv \frac{1}{2}(U_{\mathcal{T}}(0)+U_{\mathcal{T}}(1))$, giving

$$MVPF_{0\to1} = \frac{1}{1+FE_{0\to1}}; \qquad FE_{0\to1} \approx \frac{\tau_c\big(\frac{1}{2}-\tau_I\big)+\frac{\kappa}{2}}{\tau_c-\kappa}\,\frac{\Delta R_{T-1}}{\frac{1}{2}\big(U_{\mathcal{T}}(0)+U_{\mathcal{T}}(1)\big)} \tag{8}$$

C.2 Empirical Inputs

Our goal is to measure the elements of Equation 8 as steady-state annual flows. For ΔR_{T-1}, the induced realization response, we take the age-specific medium-run (2009–2013) realization responses from Figure 4 (summarized as a binary result in Appendix Table A4), which in aggregate yield \$341 million per year of additional taxable realizations before death. For $U_{\mathcal{T}}(1)$, we measure the inherited capital gains in the post-period directly: capital gains in inherited stock averaged \$875 million per year over 2009–2013. For $U_{\mathcal{T}}(0)$, we scale the pre-reform (2003–2004) flow of capital gains in inherited stock by the growth of inherited stock values between the two windows, yielding \$750 million per year. With $\tau_I = 0.10$ (the top inheritance tax rate in force during the 2009–2013 measurement window), these numbers deliver $FE = 0.20$ and an MVPF of 0.84. Appendix Table A7 collects all inputs to this calculation, and the next subsection probes the robustness of the MVPF under alternative assumptions. The conclusion that survives every variant is that the MVPF of step-up is below one: step-up actively amplifies the distortion from the capital gains tax.

Table A7: Inputs to the MVPF Calculation

Input	Value	Construction
$\Delta R_{\mathcal{T}-1}$	\$341m/yr	Medium-run (2009–13) ATT of \$213 per treated person-year (Appendix Table A4) $\times$ 1,601,469 treated individuals.
$U_{\mathcal{T}}(1)$	\$875m/yr	Average observed flow of capital gains in inherited stock (listed and unlisted), 2009–2013.
$U_{\mathcal{T}}(0)$	\$750m/yr	Average stock ICG flow from 2003–04 (\$357m), scaled 2.10$\times$ to mirror the growth of inherited stock values between 2003–04 and 2009-13.
τ_c	0.28	Statutory capital gains rate, 1992–2013.
τ_I	0.10	Top inheritance tax rate for close relatives from 2009–2013.
κ	0.02	Inheritance-tax credit per dollar of carried-over gain: carryover reduced the inheritance-tax base by 20% of the gain (Section 3), so $\kappa = 0.2\tau_I$.
$FE_{0\to1}$	0.20	$\big(\tau_c(\frac{1}{2} - \tau_I) + \frac{\kappa}{2}\big)\, \Delta R_{\mathcal{T}-1} \,/\, \big((\tau_c - \kappa)\, \frac{1}{2}(U_{\mathcal{T}}(0) + U_{\mathcal{T}}(1))\big)$
$MVPF_{0\to1}$	0.84	$1/(1 + FE_{0\to1})$

Note: This table collects our empirical inputs to the MVPF calculation.

C.3 Alternative MVPF Calculations

Our finding of an $MVPF < 1$ is stable across alternative measurement choices. Ignoring the inheritance-tax valuation deduction ($\kappa = 0$) raises the estimate from 0.84 to 0.86. Using the pre-2009 top inheritance tax rate ($\tau_I = 0.20$, with the matching deduction $\kappa = 0.04$) gives 0.85; both the deduction and the inheritance tax itself lapsed in 2014, and parameterizing that post-2014 environment ($\tau_I = \kappa = 0$) gives 0.83. Substituting for the medium-run estimates everything from the short-run era (2005–2008 responses, 2006–2008 inheritance flows, and the 20% rate then in force) gives 0.89. Removing the growth adjustment on $U_{\mathcal{T}}(0)$ entirely ($G = 1$) lowers the estimate to 0.79. Relaxing the simplifying assumptions of the derivation lowers the MVPF: if induced realizations do not displace eventually-inherited gains, the heir-side offsets disappear and the estimate falls to 0.71;[55] if the heir defers realization so that only 75% (50%) of the statutory capital gains tax is collected in present value, the estimate falls to 0.74 (0.60).

[55] Equation 8 assumes each induced dollar displaces a dollar of eventually-transferred gains, while our measured ΔR is an annual response of living owners aged 50 and above, which cannot by itself establish that mapping. Our baseline should therefore be read as conditional on full displacement. The no-offset variant prices the opposite case of no displacement (i.e., if the heirs never realize), and the MVPF remains below one everywhere between the two poles. The remaining alternative reading (that induced sales merely advance realizations that would have occurred later in life by the original owner) would imply an eventual reversal of the response, which we do not find in our event studies.

D Data Appendix

Our paper uses transactions data to build new measures of unrealized and inherited capital gain at the individual-asset level. We detail and present several validation checks on components of these novel measures below.

D.1 Detail on Sample Selection

We exclude all individual-security ledgers with transactions indicating the borrowing or lending of shares (e.g., for short selling). Affected positions are rare: 0.41 percent of person–security combinations contain at least one borrowing or lending transaction, and 0.46 percent of individuals in the stock transactions data hold at least one such position.[56] They are, however, unusually transaction-dense, averaging roughly fifteen times as many records as an unaffected position, so the exclusion removes 5.8 percent of transaction records. That density is itself evidence that these positions represent short-term trading rather than accumulated appreciation, and are therefore unlikely to contain large unrealized capital gains. Furthermore, our alternative results using an average basis calculation include these records and do not show any significant difference in any of our results.

D.2 Valuing Unlisted Firms in the Norwegian Data

Our preferred estimate of unlisted firm value largely follows the procedure of Smith et al. (2023) to compute a liquidity-adjusted, equal-weighted average of capitalized pro rata sales, assets, and EBITDA. Specifically, we apply a 10% liquidity discount to an equal-weighted average of three measures of the firm's capitalized value using (1) sales (2) assets and (3) EBITDA, respectively.

To capitalize each factor, we define valuation multiples for each NACE three-digit industry using Worldscope data on listed firms in the European Union, European Economic Area, and Switzerland.[57] We regress a ratio of the listed firm's enterprise value relative to the factor ($X \in \{sale, asset, ebitda\}$) on year-specific fixed effects for the firm's industry and country. We then employ the fitted values for Norway as our multiple for the given industry in the given year. The resulting figure is a capitalized enterprise value, to which we add the book value of the firm's non-operating financial assets and subtract the book value of its liabilities.

[56] Both shares are computed on the ledger sample of personal shareholders entering our first-in-first-out matching.

[57] Limiting ourselves to only Norwegian listed firms does not allow sufficient industry variation to estimate multipliers for all the industries of unlisted firms.

We assign unlisted firms in industries with insufficient data (fewer than 5 listed firms in the Worldscope data that year) a multiplier computed at a higher level in the industry classification (e.g., NACE 2, Sector, or the overall market average). We winsorize outlier multipliers at the 97.5th percentile. For firms that own other firms, we capitalize the firm value at the lowest level possible, and then sum over those valuations to value the parent firm. Since liquid financial assets are valued in the firm's book at market value, we value unlisted firms with substantial financial assets and little inherent value (holding companies) simply using the book values of assets minus liabilities. Finally, the equal-weighted capitalization approach can undervalue asset-heavy firms (e.g., holding companies and real estate firms) where income and sales multipliers are inappropriate. We therefore set a floor for unlisted firm value at the (book-based) tax valuation.

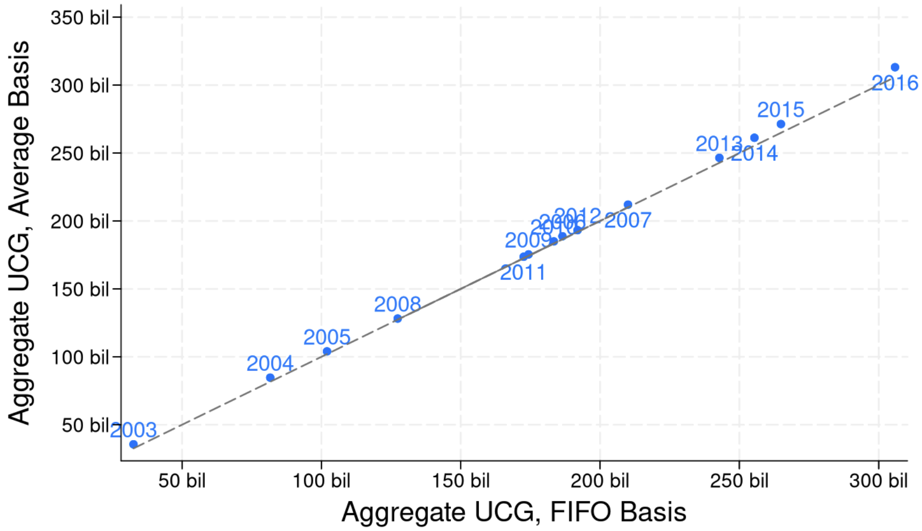
D.3 Robustness to Alternative Basis Calculation

Norway's tax code specifies that capital gains are computed on a first-in, first-out (FIFO) basis, and our primary empirical estimates accordingly employ a FIFO cost basis constructed by matching each asset disposal to its original acquisition in the transactions data. However, implementing FIFO at the transaction level requires normalizing nominal share counts across stock splits and consolidations so that acquisitions and dispositions can be matched on either side of such changes. This normalization step introduces a potential source of measurement error: if the imputed split or consolidation multiplier is incorrect for a given firm-date, the matching algorithm may fail to link the appropriate lots, and the resulting basis calculation will inherit the error. The computational burden of the FIFO matching procedure is also substantial. Additionally, in Section 4, theoretical tractability demands that we instead adopt an average basis assumption for purposes of the model.

As a robustness check, we therefore also compute unrealized capital gains under an average cost basis assumption, in which the cost basis of disposed shares equals the weighted average acquisition cost of all shares currently held. The average basis approach is both simpler to implement and computationally faster, as it requires only a running weighted average rather than lot-level matching across transactions. It also avoids the issue of stock split and consolidation adjustments. As Figure A8 demonstrates, aggregate unrealized capital gains are nearly perfectly aligned across the two methods, suggesting that the choice of cost basis rule does not materially affect our main estimates.[58]

[58] Aggregate UCG under the average basis calculation is a little larger than under FIFO, as would be expected if the "first in" positions contain higher than average gains

Figure A8: Aggregate Unrealized Capital Gains (UCG) Comparison, FIFO vs. Avg. Basis



Note: This figure compares our series for the aggregate amount of unrealized capital gains in stock calculated with our baseline first-in-first-out (FIFO) basis assumption (x-axis, in billions of 2018 USD) to the same series calculated with an average basis assumption (y-axis) from 2003 to 2018. As a point of reference, the dashed line represents perfect agreement (y=x).

D.4 Validating End-of-Year Stock Holdings

From 2004, the Norwegian Shareholder Registry documents the ownership shares by Norwegian shareholders of AS and ASA (limited liability) companies. We can therefore compare the ownership amounts documented in the Shareholder Registry data with our transaction-based calculation of individual shareholders' end-of-year holdings in these companies.

The two measures of ownership agree very closely: for AS and ASA companies subject to Shareholder Registry reporting, we are able to match 97% of Norwegian shareholders in our data into the Shareholder registry, and 99.6% of all matched person-year observations agree perfectly between the two data sources. That is, our transaction-based calculation of a given individual's number of shares owned at the end of the year in a given company usually agrees perfectly with the number of shares that the Shareholder Registry says that individual owns in that company. Residual disagreement (illustrated in Appendix Figure A9 Panel A) is probably due to issues with merging the two datasets, or limitations in our procedures to account for mergers and other company transformations.

D.5 Comparing Realized Capital Gains to Tax Returns

Our measures of unrealized and inherited capital gains are novel, and thus hard to compare to existing estimates. In contrast, realized capital gain is a component of taxable income, and is documented each year in every individual's tax return. Validating our calculation for realized capital gain allows us to conclude that our measure of basis is sensible, and that the transactions data reflects realizations in the tax return.

However, we are not able to make an exact comparison between two measures of realized capital gain in stock. The individual tax returns report total taxable capital gains and losses and real estate-specific taxable gains and losses. Below, we compare net taxable realized capital gains from assets other than real estate to our measure of realized capital gains from directly-held stock (listed and unlisted), net of the rate-of-return deduction. These measures may differ if the individual realized capital gains or losses in asset classes other than real estate or directly-held stock (e.g., from mutual funds). They may also differ (although probably only slightly) because individuals can deduct transaction costs (such as broker fees) from their taxable gain, and we do not observe these costs in our data.

Despite these known differences, the two measures of realized gain align well. 97% of all individuals in the tax data report net realized capital gain from non-real estate that is within rounding error (2000 NOK) from our calculation in the stock transactions data. Appendix Figure A9 Panel B illustrates the comparison.

D.6 Comparing Aggregate Values to Andresen (2022)

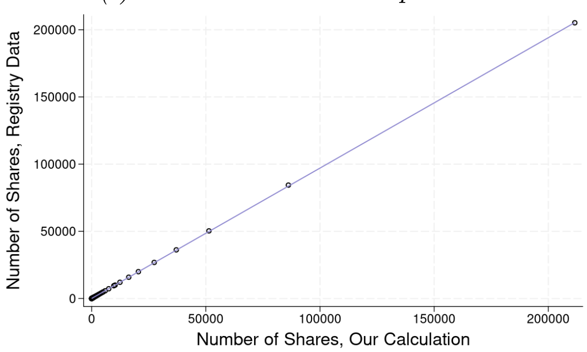
Andresen (2022) presents aggregate measures of unrealized capital gains for unlisted companies in Norway from 2006-2016. This Statistics Norway report is the only other work we are aware of that makes such a calculation. Andresen calculates unrealized capital gains as equal to the value of the unlisted stock minus basis and rate-of-return deduction. He predicts the market value of unlisted stock by multiplying book value times a book-to-price multiplier calculated from listed Norwegian companies. He has a direct measure of basis and rate-of-return deductions from the Norwegian Tax Authority (data we ourselves do not have access to), but acknowledges large problems and unlikely outliers in this measure.[59] Despite these data issues and the different procedure, his aggregate

[59]To quote from the report regarding the issue with the Norwegian Tax Authority's data on basis (not used in our own analysis): "The data on basis unfortunately has several weaknesses. In many cases, the basis values will be too large [...] Somewhat surprisingly, there is also a significant number of negative basis values in this data set, and for the years 2006-2009 this is so common that the sum of all basis values for unlisted shares is negative [...] it is unlikely

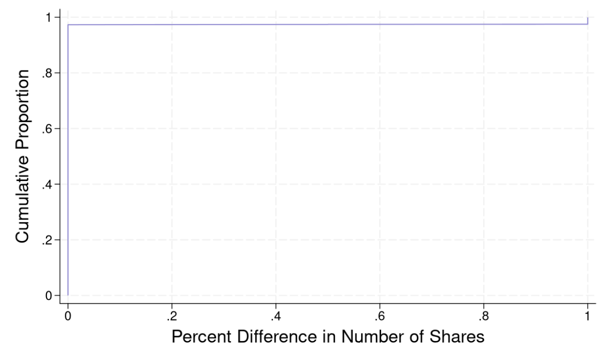
Figure A9: Validation Exercises

Panel A: End-of-Year Share Comparison

(i) Binned Scatter Comparison

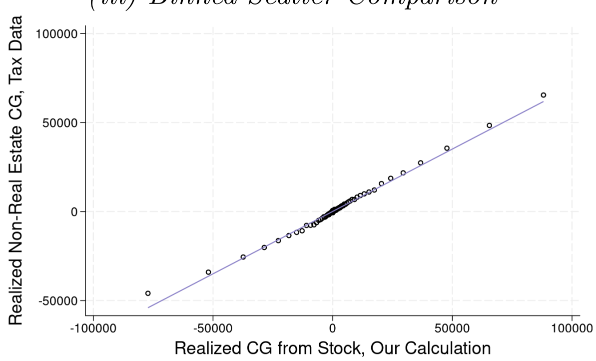


(ii) CDF of Disagreement

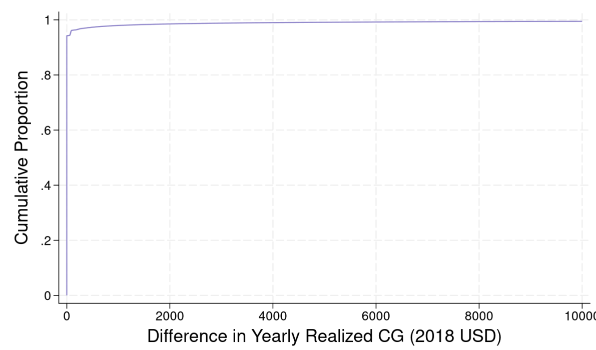


Panel B: Realized Capital Gains Comparison

(iii) Binned Scatter Comparison



(iv) CDF of Disagreement

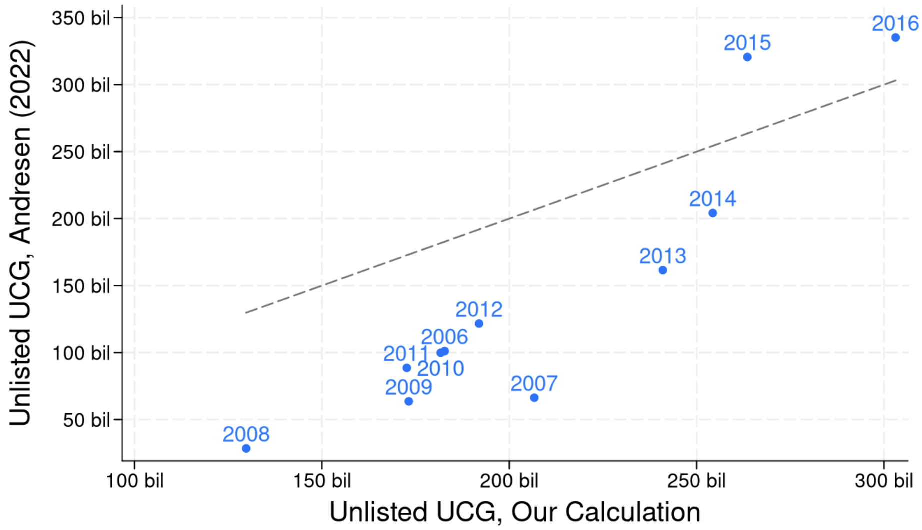


Note: Panel A compares our calculation of end-of-year shareholdings (at the individual-company-shareclass-year level) to stock ownership reported in the Shareholder Registry. This panel includes every shareholding of AS and ASA companies in our data from 2004-2018, whether or not we are able to match that shareholding into the Registry. 97% of all shareholdings match and agree exactly between the two data sources. Failures to find the shareholding (i.e., the year-person-company-shareclass record) in the Registry are coded as having zero Registry shares, and these failures to match account for the residual disagreement illustrated in Subfigure (ii). Panel B compares our calculation of individuals' yearly net realized capital gains in stock to non-real estate net realized capital gains reported in individuals' tax returns. This panel includes every person with an individual tax return from 2003-2004 and 2006-2018 (2005 is excluded due to our inability to separate out real estate realizations in the tax return data that year). 97% of all yearly realizations agree within rounding error. As depicted in Subfigure (iii), larger disagreements are associated with higher values in the tax returns than our own transaction-based calculation, consistent with the fact that the tax return-based measure includes realizations from non-real estate assets other than directly held stock.

amounts of unrealized capital gains are broadly aligned with our own estimates for unlisted firms (see Appendix Figure A10), given the acknowledged valuation and basis differences.

that these negative basis values are real." (Andresen, 2022) (authors' own translation)

Figure A10: Aggregate Unrealized Capital Gains (UCG) Comparison with Andresen (2022)



Note: This figure compares our series for the aggregate amount of unrealized capital gains in unlisted companies (x-axis, in billions of 2018 USD) to the same series from Andresen (2022) (y-axis) from 2006 to 2016. As a point of reference, the dashed line represents perfect agreement (y=x). The series from Andresen (2022) is taken from Figure 2(a) of that report, where it is reported (in 2021 kroner) as "skattepliktige latente eierinntekter" (taxable latent owner income), using the method of valuing unlisted stock with book-to-price multipliers. Values are then converted into 2018 dollars for comparison with our estimates. It is worth noting that Andresen (2022) expresses substantial reservations about the quality of his data on basis, especially for years 2006-2009. Given the data and methodological differences between our approaches (detailed in Appendix Section D.6), we consider the depicted level of agreement to be reasonable.

E Benchmarking Our Realization Response Against Rate Elasticities

The removal of step-up raised taxable realizations without changing the statutory capital gains tax rate. To gauge the magnitude of this response, we ask: how large a cut in the capital gains tax rate would have been required to generate a realization response of the same size, given realization elasticities estimated in the prior literature? This appendix formalizes that conversion.

Setup

Let $R(\tau)$ denote aggregate realized capital gain as a function of the capital gains tax rate τ, and let $\tau_0 = 0.28$ be the Norwegian capital gains tax rate throughout our baseline period. Our reduced-form estimates imply that the removal of step-up increased taxable realizations by $g = 26\%$ relative to counterfactual.[60] The rate τ^{eq} that would deliver an equivalent response solves $R(\tau^{\text{eq}})/R(\tau_0) =$

[60] This is the aggregate response implied by our baseline (age-based) design over 2005–2008, expressed relative to the counterfactual base (Section 6.2). Because the estimated response is stable across our post-period, with no decay

$1 + g$, which requires specifying the realization function $R(\tau)$ over the interval between τ^{eq} and τ_0. Rather than impose a single functional form, we take two recent elasticity estimates from the literature together with the (different) functional forms used to create each estimate.

Our first benchmark, Agersnap and Zidar (2021), estimates a constant elasticity of realizations with respect to the net-of-tax rate:

$$R(\tau) \;=\; R(\tau_0)\left(\frac{1-\tau}{1-\tau_0}\right)^{\varepsilon}, \tag{9}$$

under which the equivalent rate is

$$\tau^{\text{eq}} \;=\; 1-(1-\tau_0)\,(1+g)^{1/\varepsilon}. \tag{10}$$

Our second benchmark, Dowd and McClelland (2024), estimates a semi-logarithmic form:

$$\ln R(\tau) \;=\; \ln R(\tau_0) \;-\; b\,(\tau-\tau_0), \qquad b \;>\; 0, \tag{11}$$

under which the equivalent rate is

$$\tau^{\text{eq}} \;=\; \tau_0 \;-\; \frac{\ln(1+g)}{b}. \tag{12}$$

To keep magnitudes comparable across the two benchmarks despite the different forms, Table A8 also reports each calibration's implied elasticity with respect to the tax rate evaluated at the common Norwegian rate $\tau_0 = 0.28$.

Choice of Elasticities

For Agersnap and Zidar (2021) we use their preferred log-log estimates directly. Their long-run (post-reform years 6–10) estimates across three specifications are $\varepsilon = 1.47$ (baseline), 1.18 (controlling for other state tax changes), and 0.99 (identifying only from reforms larger than one percentage point). At $\tau_0 = 0.28$, these imply local tax-rate elasticities of 0.39 to 0.57.

We also use the long-horizon (four years of lags) specification of Dowd and McClelland (2024) (an update of Dowd et al. (2015)). Because their model is semi-log, their permanent elasticity of -0.59 is evaluated at their sample's rates and must be restated at the Norwegian rate: dividing

through 2018, we interpret g as the persistent annual response over our observed horizon and accordingly compare to long-run elasticity estimates.

by the mean marginal tax rate of 17.4 percent among realizers in the same panel recovers the coefficient $b = 0.59/0.174 = 3.39$, implying a local elasticity of 0.95 at $\tau_0 = 0.28$.[61]

It is worth noting that Gravelle (1991, 2026) argues that permanent elasticities above roughly 0.34 are infeasible, which would rule out the larger estimates here. Adopting that bound would only strengthen our conclusion: a smaller elasticity implies a larger rate cut is needed to match our estimated response, and hence a larger revenue loss from the equivalent rate-cut policy.

Results

Table A8 reports τ^{eq} from Equations (10) and (12). Matching the aggregate realization response of 26 percent would have required cutting the Norwegian capital gains tax rate from 28 percent to 21.2 percent under the Dowd–McClelland benchmark, and to between 15.7 and 9.1 percent under the Agersnap–Zidar estimates—that is, eliminating between roughly a quarter and more than two thirds of the rate.

These comparisons also imply an immediate contrast in revenue expectations. Since $R(\tau^{\text{eq}})/R(\tau_0) = 1 + g$ by construction, capital gains tax revenue at the equivalent rate, relative to revenue at τ_0, is

$$\frac{\tau^{\text{eq}}\,R(\tau^{\text{eq}})}{\tau_0\,R(\tau_0)} \;=\; \frac{\tau^{\text{eq}}}{\tau_0}\,(1+g) \tag{13}$$

under either functional form. Whether this ratio falls below one depends on the revenue-maximizing rate, which both studies' baseline estimates put above the Norwegian rate of 28 percent. Under either paper's implied Laffer curve, the equivalent rate cut would therefore have reduced capital gains tax revenue (by 5 to 59 percent across the calibrations in Table A8), whereas the removal of step-up generated the same realization response while raising revenue by 26 percent. The economic content of this contrast does not depend on the exact revenue figures: a rate cut necessarily lowers the rate on all inframarginal realizations, while removing step-up can raise realizations while maintaining a stable tax rate.

[61] Dowd and McClelland (2024) do not report the exact rate at which their elasticity is evaluated; we use the weighted mean marginal rate among realizers reported for the same 1999–2008 panel in Dowd et al. (2015). Note that restating a 28-percent-rate elasticity this way assumes the linear dose-response of Equation 11 holds between the sample mean and 28 percent; combined federal-plus-state rates of 28 percent are within the support of their individual-level data.

Table A8: Rate Cut Required to Match the Realization Response to the Removal of Step-Up

Calibration	Functional form	Elasticity at $\tau_0 = 0.28$	Equivalent rate τ^{eq}	Implied cut (pp.)	Revenue change
A–Z, baseline	log-log ($\varepsilon = 1.47$)	0.57	15.7%	12.3	−29%
A–Z, other-tax controls	log-log ($\varepsilon = 1.18$)	0.46	12.4%	15.6	−44%
A–Z, large reforms only	log-log ($\varepsilon = 0.99$)	0.39	9.1%	18.9	−59%
Dowd–McClelland (2024)	semi-log ($b = 3.39$)	0.95	21.2%	6.8	−5%

Note: This table reports the capital gains tax rate τ^{eq} that would generate a realization response equal to our estimated response to the removal of step-up ($g = 0.26$). Estimates from Agersnap and Zidar (2021) ("A–Z") are taken from the post-reform years 6–10 rows of their Table 2. For Dowd and McClelland (2024), we take their four-lag permanent elasticity of -0.59 evaluated at the mean marginal rate of 17.4 percent among realizers in the same panel (Dowd et al., 2015), $b = 0.59/0.174 = 3.39$. The elasticity column reports the implied local elasticity of realizations with respect to the tax rate at the Norwegian baseline rate $\tau_0 = 0.28$: $\varepsilon\,\tau_0/(1-\tau_0)$ for the log-log rows and $b\,\tau_0$ for the semi-log row. The equivalent rates τ^{eq} are estimated from Equation 10 and Equation 12. Revenue changes are computed from Equation 13 and refer to capital gains tax revenue only.

Caveats

Three caveats apply. First, these elasticities are identified from U.S. variation, either state-level tax changes (Agersnap and Zidar, 2021) or individual federal-plus-state rate variation (Dowd and McClelland, 2024); portability to Norway is imperfect. Second, the required cuts lie below the support of the identifying variation in these studies, so τ^{eq} should be read as an order-of-magnitude benchmark rather than a point prediction. We mitigate this concern by evaluating each estimate under its own functional form and by showing that the functional-form choice does not overturn the conclusions. Third, the comparison holds fixed the composition of realizations, which we might expect to be differently affected by the two policy levers: in particular, a rate cut lowers the price of all realizations while step-up only affects the return to deferring gains until death.

F Background on Norway

F.1 Taxation in Norway, 1992-2018

Norwegians pay taxes on their worldwide labor income, capital income, and wealth. Labor income is subject to progressive taxation with a top marginal rate of 53.2–54.3% (including social security contributions) during our period of study. Net wealth in excess of an exemption amount faces a yearly tax of 0.85–1.30% during our period.[62]

[62] The wealth tax exemption has ranged from $19,555 to $162,800 (2018 USD) during our period of study. In general, exemptions have grown more generous and wealth tax rates have fallen over our period of study. See Thoresen et al. (2022) for a detailed description of Norway's wealth tax. Interestingly, Norway collects a smaller share of total tax

F.1.1 Capital Gains and Inheritance Taxation

Norway has a flat capital gains tax on realized capital gains. From 1992 to 2013, all (net) realized capital gains were taxed at a rate of 28%.[63] Since 2014, there have been several rate changes, including the introduction of separate rates for the realization of capital gains from stock and non-stock assets. These rate changes are depicted in Figure A1. Throughout our study period, the capital gains tax has had a flat rate structure, so that capital gains tax rates do not depend on individual income.

Prior to its abolition in 2014, Norway's integrated gift and inheritance transfer tax taxed the receipt of gifted or inherited assets through a rate structure that depended on the cumulative amount of the gifts/inheritance received and the relationship between the giver and the recipient. Other than a small yearly gift exemption, no distinction in tax rates was made between *inter vivos* gifts and bequests at the time of death. Appendix Table A2 provides details on the rates and brackets over time. Figure A1 depicts the top rate for gifts and inheritances between close relatives: a rate that fell from 20% to 10% in 2009.[64] In 2014, the gift and inheritance tax was abolished, so all rates for subsequent years are set to 0%.

F.1.2 Stepped-Up Basis Policy

During our period of study, Norway moved from a system of stepped-up basis for inherited capital gains to a system of carryover basis. This change was accomplished in two steps. In 2006, carryover basis was introduced for intergenerational transfers of appreciated stock (including equity-based mutual funds). Recipients of gifted or inherited stock from 2006-2013 continued to be subject to inheritance tax on value of the stock they received, but now were also subject, if and when they sold the stock, to capital gains tax on the amount of appreciation during the original owner's holding period. Losses, however, were not permitted to be carried over in the 2006 reform – so stocks with losses continued to be subject to a "step-down" in basis. In 2014, concurrent with the abolition of the inheritance tax, carryover basis was introduced for other assets, including stocks with losses. Therefore, since 2014, recipients of intergenerational transfers do not pay inheritance tax at the time of the receipt, but are subject to capital gains tax on appreciation from the original owner's

revenue from property and wealth taxation than is typical among OECD countries (OECD, 2018).

[63] This rate is also applied to other forms of capital income, such as interest income. Capital losses can be used to fully offset capital gains and ordinary income (but not social security contributions or top marginal tax rates).

[64] The definition of close relatives includes parents and children of the giver. Transfers between spouses were not taxed. Transfers between individuals who are not close relatives were subject to a higher top rate in every year: 30% before 2009, and 15% until 2014.

holding period when they sell the inherited asset.

Certain types of real estate are exempt from capital gains taxation in Norway if the owner's use of the property and length of ownership meet certain requirements. Most prominently, residential property is not subject to capital gains tax if the seller has owned and lived in the property as his primary residence for 12 of the last 24 months. Vacation homes can be similarly exempt if the seller has owned the property for more than five years and used it as his own vacation home for at least five of the last eight years. Buildings on a farm may be exempt if the farm has been owned for over ten years. In these cases in which the original owner could have sold the property without capital gains tax, the heir's basis continues to be stepped up to the assumed sales value at the time of acquisition, even after 2014.

F.1.3 Legislative History & Anticipation of Step-Up's Removal

Norway's policy of stepped-up basis for intergenerational transfers was first established by Norwegian Supreme Court rulings in the 1920s. Various government committees sporadically proposed moving to a system of carryover basis (proposals were made in 1975, 1981 (for gifts only), 1989 and 2000), but no changes were enacted until 2006 (Zimmer, 2014).

In 2005, it was announced that step-up would be removed for stock (including unlisted stock and equity-based mutual funds).[65] The 2006 removal of stepped-up basis for inherited stock was passed as an afterthought to a wider reform of capital taxation. The 2006 tax reform changed the exemption rules for dividend and capital gains taxes in a way that made these exemptions more dependent on owners' cost basis (see Section F.1.4). Because of concern that owners might transfer shares as gifts to their relatives to get the increased dividend exemptions that would accompany stepped-up basis, subsequent legislation was passed that abolished stepped-up basis for shares starting in 2006. Although the main components of the 2006 tax reform were formally announced in March 2004 and approved in June of that year, the removal of step-up was not part of the June legislation and was not definitively decided until later in the year (Norwegian Ministry of Finance, 2004).

[65] The removal of step-up for stock was formally introduced on December 10, 2004 (Stortinget, 2004). However, this change did not receive immediate media attention, so we take 2005 to be our announcement year. A search through the Norwegian media library (Nasjonalbiblioteket) for the terms "kontinuitetsprinsippet" and "diskontinuitet" (which are the Norwegian terms corresponding to carryover and stepped-up basis, respectively) shows that the first mention of the impending change appeared in the newspaper *Finansavisen* on January 8, 2005, under the headline "One Year Left" (Amundsen, 2005). A second mention appeared in *Levanger-Avisa* on February 19, 2005 (Vigdal, 2005). Both articles featured lawyers advocating that Norwegians transfer stock (especially stock of family businesses) to the younger generation before the new carryover basis regime came into effect. Media coverage of the upcoming change grew more intense during the spring and summer of 2005.

The 2014 removal of step-up for other assets was closely tied to the abolition of the inheritance tax. Right-wing parties campaigned on abolishing the inheritance tax during the 2013 election, which they won in September 2013. Therefore from September 2013, it was reasonably certain that the inheritance tax would be repealed, although the exact timing of the repeal initially remained uncertain. The rules accompanying that repeal, which included the removal of step-up for non-stock assets, were announced as part of the government's yearly budget in October 2013.

F.1.4 Other Aspects of the 2006 Reform

As mentioned above, the removal of step-up for stock came about due to a larger overhaul of exemptions for capital gains and dividend taxation. The overall reform was primarily motivated by a need to prevent private business owners from relabeling labor income as dividends, which had become endemic to the old system. Under the old system (called "RISK"), both capital gains and dividends were given generous exemptions that corresponded to the amount of retained earnings held in the underlying firm. The goal of this exemption was to avoid double-taxation on corporate earnings which would have already been subject to the corporate tax. Furthermore, dividend distributions were subject to a yearly limit that corresponded to the amount of retained earnings in the firm at the beginning of the year. Because the exemption amount and dividend cap were thus linked, the effect was that almost no dividends were taxed prior to 2006. From 2006, the same dividend cap remained in place, but the exemption calculations for dividends and capital gains were made less generous going forward. Henceforth, investors would accumulate exemptions based on how their cost basis would grow with the normal rate of return.

Previous work has documented that the 2006 reform led to a large fall in dividend distributions (Alstadsæter and Fjærli, 2009; Alstadsæter et al., 2014, 2019). However, we do not think that this aspect of the reform is likely to be linked to the increase in capital gains realizations we document in Section 5. First of all, our results in Section 5 show a response starting in the announcement year of 2005, before the exemption calculations changed.[66] Secondly, the old exemption amounts remained usable for realizations after 2006 – only the accumulation of future exemption amounts changed.[67] Therefore, there was no immediate change to the tax calculation of realizing capital gains in 2005 vs. early 2006 – the same exemption amounts would be available both times. Finally,

[66]It is important to recall that our realization results look at the outcome of *taxable* realized capital gains, and so do not include any (tax-free) company transformations occurring under Transition Rule E.

[67]See The Norwegian Tax Administration (n.d.): "For shares acquired before 1 January 2006, the input value is adjusted for accumulated RISK."

we do not think there is any incentive for investors to shift from dividend distributions to capital gains realizations. Both before and after the reform, dividends and capital gains were subject to the same tax rate and granted the same exemption amounts. Investors should therefore be "neutral" between the two methods of extracting money from the firm both before and after the reform (Sørensen, 2005).[68] We therefore do not expect our documented increase in capital gains realizations (Section 5) to be a shift from dividends to capital gains realizations.

Appendix References

Agersnap, Ole and Owen Zidar, "The Tax Elasticity of Capital Gains and Revenue-Maximizing Rates," *American Economic Review: Insights*, 2021, *3* (4), 399–416.

Alstadsæter, Annette and Erik Fjærli, "Neutral Taxation of Shareholder Income? Corporate Responses to an Announced Dividend Tax," *International Tax and Public Finance*, 2009, *16*, 571–604.

_ , Wojciech Kopczuk, and Kjetil Telle, "Are Closely-Held Firms Tax Shelters?," *Tax Policy and the Economy*, 2014, *28*, 1–32.

_ , _ , and _ , "Social Networks and Tax Avoidance: Evidence from a Well-Defined Norwegian Tax Shelter," *International Tax and Public Finance*, 2019, *26*, 1291–1328.

Amundsen, Toril Flate, "Ett år Igjen [One Year Left]," *Finansavisen*, 2005.

Andresen, Martin Eckhoff, "Latente eierinntekter og skatteplikt på unoterte aksjer etter aksjonærmodellen [Latent Owner Income and Tax Liability on Unlisted Shares according to the Shareholder Model]," Reports 2022/37, Statistics Norway 2022.

Bjerksund, Petter and Guttorm Schjelderup, "Aksjonærmodellen og fritaksmetoden: Et to-hodet troll? [The Shareholder Model and the Exemption Method: A Two-headed Troll?]," *Samfunnsøkonomen*, 2021, *135* (4), 53–63.

_ and _ , "Er den norske aksjonærbeskatningen nøytral? [Is Norwegian Shareholder Taxation Neutral?]," *Samfunnsøkonomen*, 2021, *135* (4), 43–52.

[68] See Bjerksund and Schjelderup (2021a,b) for discussions of how exemptions and loss limitations affect the capital gains realization decision in Norway.

Dammon, Robert M., Chester S. Spatt, and Harold H. Zhang, "Optimal Consumption and Investment with Capital Gains Taxes," *The Review of Financial Studies*, 2001, *14* (3), 583–616.

Dimson, Elroy, Paul Marsh, and Mike Staunton, "The Worldwide Equity Premium: A Smaller Puzzle," in Rajnish Mehra, ed., *Handbook of the Equity Risk Premium*, Elsevier, 2008.

Dowd, Tim and Robert McClelland, "The Sensitivity of the Tax Elasticity of Capital Gains to Lagged Tax Rates and Migration," 2024.

_ , _ , and Athiphat Muthitacharoen, "New Evidence on the Tax Elasticity of Capital Gains," *National Tax Journal*, 2015, *68* (3), 511–544.

Eika, Lasse, Magne Mogstad, and Ola L. Vestad, "What Can We Learn about Household Consumption Expenditure from Data on Income and Assets?," *Journal of Public Economics*, 2020, *189*, 104163.

Fagereng, Andreas, Charles Gottlieb, and Luigi Guiso, "Asset Market Participation and Portfolio Choice over the Life-Cycle," *The Journal of Finance*, 2017, *72* (2), 705–750.

Gravelle, Jane G., "Limits to Capital Gains Feedback Effects," *Tax Notes*, 1991.

_ , "Capital Gains Tax Options: Behavioral Responses and Revenue," *Congressional Research Service Report*, 2026.

Grytten, Ola H., "A Consumer Price Index for Norway, 1516–2003," in Øyvind Eitrheim, Jan Tore Klovland, and Jan Fredrik Qvigstad, eds., *Historical Monetary Statistics for Norway 1819–2003*, Oslo: Norges Bank, 2004.

Kleven, Henrik J., "Sufficient Statistics Revisited," *Annual Review of Economics*, 2021, *13* (1), 515–538.

Klovland, Jan Tore, "Bond Markets and Bond Yields in Norway 1820–2003," in Øyvind Eitrheim, Jan Tore Klovland, and Jan Fredrik Qvigstad, eds., *Historical Monetary Statistics for Norway 1819–2003*, number 35, Oslo: Norges Bank, 2004, pp. 99–180.

Norwegian Ministry of Finance, "Skattereformen 2006 [Tax Reform 2006]," 2004. Accessed: 2024-10-14.

Ødegaard, Bernt Arne, "Asset Pricing at the Oslo Stock Exchange: A Source Book," Working paper 2007.

OECD, "The Role and Design of Net Wealth Taxes in the OECD," *OECD Tax Policy Studies*, 2018, (26).

Smith, Matthew, Owen Zidar, and Eric Zwick, "Top Wealth in America: New Estimates under Heterogeneous Returns," *The Quarterly Journal of Economics*, 2023, *138* (1), 515–573.

Sørensen, Peter Birch, "Neutral Taxation of Shareholder Income," *International Tax and Public Finance*, 2005, *12* (6), 777–801.

Statistics Norway, "StatBank Table 07902: Life tables, by sex and age," 1966–2024. Accessed 11 November 2025.

Stortinget, "Innst. O. nr. 10 (2004–2005): Innstilling fra finanskomiteen om skatte- og avgiftsopplegget 2005 – lovendringer [Recommendation from the Finance Committee on the Tax and Levy System 2005 – Legislative Changes]," 2004. Kildedokument: Ot.prp. nr. 1 (2004–2005); 19 November 2004.

The Norwegian Tax Administration, "Aksjonærmodellen [The Shareholder Model]," n.d. Accessed: 6 June 2025.

Thoresen, Thor O., Marius A. K. Ring, Odd E. Nygård, and Jon Epland, "A Wealth Tax at Work," *CESifo Economic Studies*, 2022, *68* (4), 321–361.

Vigdal, Anne Marit, "Generasjonsskifte i Næring [Generational Succession in Business]," *Levanger-Avisa*, 2005. 19 February 2005.

Zimmer, Frederik, "Generation Shifting and the Principle of Continuity in Norwegian Tax Law," *Nordic Tax Journal*, 2014, *1*, 93–101.